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Conceptual design and analysis of a photon calibration system for ETpathfinder

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Abstract

This thesis explores the design and analysis of a photon calibration system for ETpathfinder, a prototype facility for the next generation of gravitational-wave detectors. The system's dominant error sources are identified, and their expected contribution to the calibration uncertainty is calculated. Ultimately, two conceptual system designs are proposed: a cost-effective single-beam photon calibrator with limited performance, and a dual-beam photon calibrator with cryogenic windows that meets the desired accuracy across the full bandwidth. The first concept serves as a cost-effective system that uses a single beam to actuate the test mass. Numerical simulations have shown that local deformation of the test mass surface can introduce significant calibration uncertainty at high frequencies, thereby limiting this system's operating bandwidth. The second concept is expected to achieve sub-percent calibration accuracy across a 10 Hz to 1 kHz modulation bandwidth. It requires a 40 mW laser at 1460 nm, closed-loop modulation, and two parallel beams to mitigate errors arising from local deformation. To provide sufficient optical access for these beams, optical windows are required in the radiation shields surrounding the test mass.

Thermal analyses of the cryogenic system have identified fused silica as a functional window material. To effectively dissipate the radiative heat from the environment, three windows are placed in the innermost shields. To ensure sufficient thermal contact with the conductively cooled shields, it is suggested to mount the windows with an intermediate indium layer. The analyses on the cryogenic windows and the conceptual designs of the photon calibration system provide a foundation for future work at ETpathfinder and the Einstein Telescope.

Contents

1 Introduction

1.1 Gravitational wave detection

Gravitational waves are propagating distortions in the fabric of space-time. They are the result of immense astrophysical events, such as the collapse of massive stars or the mergers of black holes or neutron stars. The existence of gravitational waves (GWs) was predicted by Einstein's general theory of relativity in 1916. Roughly a hundred years later, in 2015, the first direct detection of a GW was made by the Laser Interferometer Gravitational-Wave Observatory (LIGO) in the United States [1].

Compared to electromagnetic radiation, GWs interact very weakly with matter and can therefore carry information from more distant regions, i.e., from earlier times of our universe. This weak interaction of compressing and stretching space-time also makes them extremely hard to detect. Typical amplitudes of a GW-induced strain are on the order of $10^{-21}\,\frac{\mathrm{m}}{\mathrm{m}}$. For a detector with 4-kilometre arms such as LIGO, this corresponds to measuring a distance smaller than one-thousandth the diameter of a proton.

1.1.1 Current detectors

The first GW observation by LIGO in 2015 has since advanced its development and that of other detectors: Virgo in Italy [2] and KAGRA in Japan [3], all of which are categorised as the second-generation GW observatories. These detectors are essentially large interferometers with kilometre-long arms, as shown in Figure 1.1. A passing GW will slightly compress or stretch one arm relative to the other, resulting in a change in the interferometer output signal. The mirrors, also referred to as *test masses*, are suspended and stabilised by vibration isolation towers, while the entire detector, including its arms, operates in ultra-high vacuum.

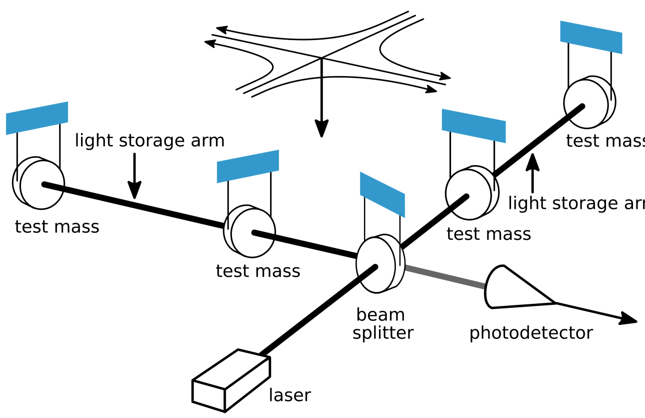


Figure 1.1: Schematic of an interferometric gravitational wave detector [4]

The LIGO and Virgo detectors operate at room temperature and use fused silica test masses. The KAGRA detector, on the other hand, operates at cryogenic conditions with sapphire test masses. Nowadays, the network of second-generation GW detectors – LIGO, Virgo, and KAGRA (LVK) – is able to detect 2 to 3 GW signals per week [5].

1.1.2 Einstein Telescope

The Einstein Telescope (ET) is projected to be the first third-generation GW detector to be built, and is expected to observe hundreds of thousands to millions of events per year [6]. To achieve this, the sensitivity of ET will need to be improved by a factor of 10 relative to the current LVK network through several design alterations. The ET interferometer arms will be 10 kilometres long compared to the 3 or 4 kilometres of current detectors, increasing the absolute strain generated by gravitational waves.

ET will likely be built in a triangular configuration rather than the conventional L-shape to achieve greater directional coverage. Each corner will house both a high-frequency (HF) and a low-frequency (LF) detector, each operating at a different temperature and optimised for sensitivity at different frequencies to broaden the detector's overall bandwidth. ET will be located 250 meters underground to mitigate seismic noise. The HF detector will operate at room temperature with fused silica test masses, while the LF detectors will use silicon test masses, cooled to approximately 15 K to reduce the thermal noise.

1.1.3 ETpathfinder

Research and development of the technologies required for the ET-LF detectors are carried out at ETpathfinder, a prototype interferometer designed to test these technologies in an ET-like environment [7]. ETpathfinder contains two folded interferometers with parallel-arm cavities and 150 mm-diameter silicon test masses. One interferometer, located in the X-arm, is operated at 15 K with a laser wavelength of 1550 nm, while the other interferometer, located in the Y-arm, is operated at 123 K with a wavelength of 2090 nm.

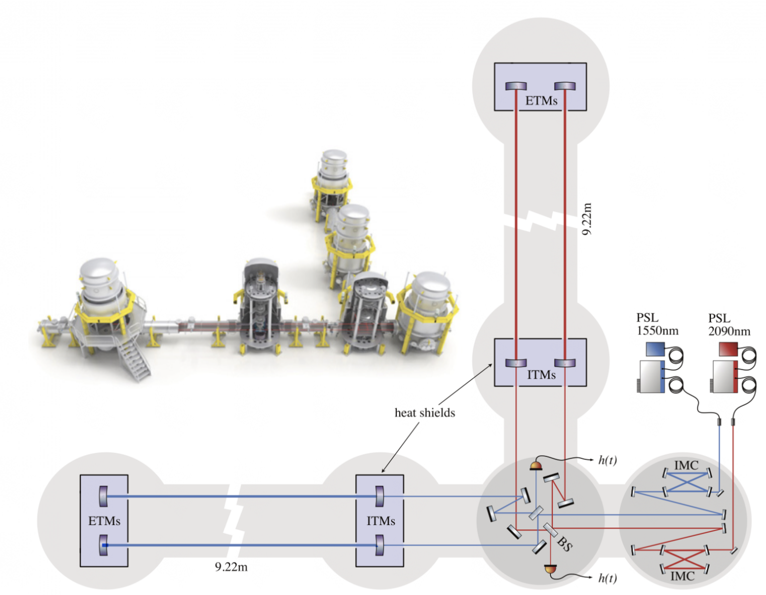


Figure 1.2: Layout of ETpathfinder with parallel arm cavities [7]

Maintaining the test masses at these cryogenic temperatures is partly achieved using thermal radiation shields. The conductively cooled shields enclose the test masses and significantly reduce the radiative heat load from the environment. The desired bandwidth for ETpathfinder ranges from 10 Hz to a few kHz, as these frequencies are similar to those of the gravitational waves to be measured by the Einstein Telescope.

1.2 Calibration of the absolute strain

To obtain accurate absolute measurements of the astrophysical events that produce gravitational waves, precise calibration of the detectors is essential. Calibration uncertainties propagate into errors in the absolute GW signal and determine the uncertainties of the inferred astrophysical parameters, such as the distance to and size of the source [8]. Calibration relates the output of the interferometer's photodiode to the absolute displacement of the test mass and, thereby, to the GW-induced strain. This can be done by exerting a known periodic force on the test mass for which the displacement of the test mass is precisely modelled, and by measuring the corresponding output signal to the interferometer.

The process of calibrating the strain amplitude of a gravitational wave detector can be summarised by the following steps:

1. **Actuate the test mass by a known periodic displacement**
2. Measure the interferometer photodiode output signal
3. Derive the transfer function from voltage to absolute strain
4. Repeat 1-3 continuously at various frequencies during operation of the interferometer

This thesis will focus on the first step in this process: actuating one of the test masses by a known displacement to generate a signal that can be measured by the interferometer. The amplitude of these displacements must be on the order of 10^{-16} m over a frequency range from 10 Hz to a few kHz. A mature and widely used method to achieve such small amplitudes is photon calibration.

1.2.1 Photon calibrator

A photon calibrator (Pcal) relies on the phenomenon of radiation pressure. When light is reflected off a mirror, its momentum is transferred to the mirror, resulting in an extremely small force on the mirror. The Pcal uses an auxiliary power-modulated laser to apply a periodic force to the test mass by reflecting the laser off its surface. This periodic excitation of the test mass is measured by the interferometer photodiode and used to calibrate the differential arm length.

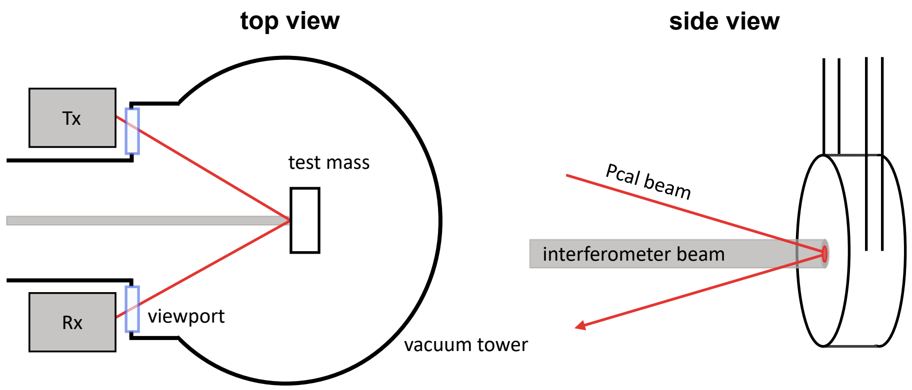


Figure 1.3: Schematic of a photon calibration system

The output power of the Pcal beam is measured at the transmitter (Tx) and receiver (Rx) modules, located outside the vacuum envelope. This is done using precisely calibrated photodiodes to obtain an accurate measurement of the reflected power. The reflected power is directly proportional to the induced displacement, serving as an independent reference for the interferometer's differential arm length. In addition to calibration, the Pcal beam can be used to inject an artificial GW signal into the detector to test the detection pipelines.

1.2.2 Calibration uncertainty

The calibration uncertainty quantifies the error in the estimation of the gravitational wave strain and, consequently, in the relative mirror displacement. Lindblom showed in [8] that the calibration uncertainty of gravitational wave detectors must be less than 0.5 % to determine the astrophysical features of a GW event with sufficient detail. For a photon calibrator, the measurement uncertainty of the photodiodes, laser power stability, and other factors contribute to the total calibration uncertainty.

1.3 Thesis objective

The Einstein Telescope will use photon calibration as one of its methods to calibrate the absolute strain. Before that happens, a prototype Pcal system will be implemented at the ETpathfinder test facility.

The implementation of a photon calibrator is not included in the initial interferometer design, as it is not a prioritised technology for research. Nonetheless, it would be a valuable addition to the prototype. A crucial aspect of implementing a Pcal system at ETpathfinder is the presence of cryogenic shields that restrict access to the test mass. This thesis explores the feasibility of implementing a photon calibrator for ETpathfinder, with a provisional requirement on the calibration uncertainty. This leads to the following research goal:

Conceptualise a photon calibration system for ETpathfinder with a calibration uncertainty of less than 1 % across a 10 Hz to 1 kHz bandwidth.

1.4 Thesis outline

In chapter 2, an overview is given on the operating principles of a gravitational wave detector and the dominant noise sources for the ETpathfinder prototype. It is further explained how a photon calibrator displaces the test mass and how this is used to calibrate the absolute arm-length variations. Then, in chapter 3, a photon calibrator concept is proposed for ETpathfinder that uses an already-present auxiliary laser, which also serves as a Pcal beam. By identifying and quantifying the dominant error sources, the expected performance of this concept is determined. In chapter 4, several other concepts are explored with the aim of further reducing the calibration uncertainty. One of these concepts is selected, and the expected calibration uncertainty is determined. The selected concept requires optical windows in the radiation shields that surround the test mass. In chapter 5, thermal analysis of the cryogenic system is conducted to determine the feasibility of such cryogenic windows. Several window materials and configurations are analysed, of which the most promising are selected. In chapter 6, the final conclusions and recommendations are presented.

2 Background and theory

In this chapter, the operating principles of a gravitational wave detector and the dominant noise sources are presented. It is explained how the absolute strain amplitude is calibrated and how radiation pressure is used to actuate the test mass.

2.1 Interferometric gravitational wave detection

The basis of a gravitational wave detector is the *Michelson interferometer*, which measures the change in the relative position between the end test masses (ETM). A laser beam is split into two beams travelling along perpendicular arms, reflected by mirrors at the end, and recombined to produce an interference signal. The sensitivity of such a detector is proportional to the arm length, but the physical arm lengths are limited to a few kilometres due to practical considerations [9]. For a GW-induced strain of $h \sim 10^{-21}$ and arm lengths of 3 km, the optical path length variations are on the order of $\Delta L \sim 3 \cdot 10^{-18}$ m, around a thousandth of the diameter of a proton.

To increase the effective optical path length, the light is folded multiple times using an additional input test mass (ITM) that reflects part of the light back, creating a Fabry-Perot cavity. This effect of folding the beam path is effective only near the cavity's precise resonance, and the effective number of round trips is determined by the finesse. The process of finding and holding the arm sufficiently close to resonance is called locking. Both arms of the gravitational wave detector contain a cavity, making the system a *Fabry–Perot Michelson interferometer* (FPMI). The laser wavelength is tuned to lock the first cavity, while the length of the second cavity is locked to the wavelength. This locking signal for the second cavity gives the relative length variation between the two arms [10].

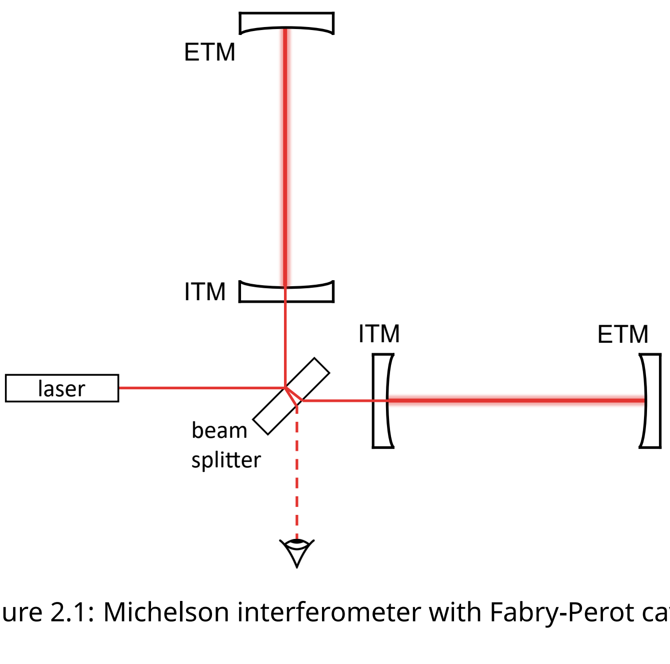


Figure 2.1: Michelson interferometer with Fabry-Perot cavities

At low frequencies, feedback control systems keep the mirrors aligned near the operating point of the interferometer, while at higher frequencies, the mirrors are sufficiently free to respond to gravitational waves

In the example where the simple Michelson configuration produces an optical path length variation of $\Delta L \sim 3 \cdot 10^{-18}$ m, the addition of Fabry–Perot cavities can increase this response to $\Delta L \sim 10^{-16}$ m, depending on the finesse of the cavity. This distance is still smaller than the thermal vibrations of individual atoms, but it is measurable because the average position of all atoms across the laser beam spot varies less than the GW-induced displacement.

2.1.1 Sensitivity

The sensitivity of a gravitational wave detector is governed by its noise level, which is usually expressed as the amplitude spectral density (ASD) $\tilde{N}(f) = \sqrt{S_N(f)}$, the square root of the power spectral density (PSD) $S_N(f)$. This choice is natural because gravitational wave detectors measure strain amplitude h, a dimensionless quantity that scales with $h \propto 1/r$, where r is the distance to the GW source. In contrast, the intensity I of an electromagnetic wave scales with $I \propto 1/r^2$ and is therefore generally characterised by a PSD.

The amplitude spectral density of a noise source is expressed in units of $\frac{\text{noise amplitude}}{\sqrt{\text{Hz}}}$, which makes it convenient to compare directly to the signal amplitudes such as strain or displacement. As an example, if a displacement signal of $\Delta L = 10^{-17}$ m is detected in a $\Delta f = 1$ kHz bandwidth, the ASD of the displacement is $\Delta\tilde{L} = \frac{\Delta L}{\sqrt{\Delta f}} = 3 \cdot 10^{-19}$ m/$\sqrt{\text{Hz}}$.

Noise sources

Noise sources of a GW detector arise from fundamental physical limits, such as shot noise and thermal noise, as well as from technical sources, such as seismic noise and laser power fluctuations. All sources are assumed to be uncorrelated and thus add in quadrature, resulting in the *sensitivity curve* of a detector. Figure 2.2 shows the expected displacement sensitivity curve for ETpathfinder together with the contributing sources.

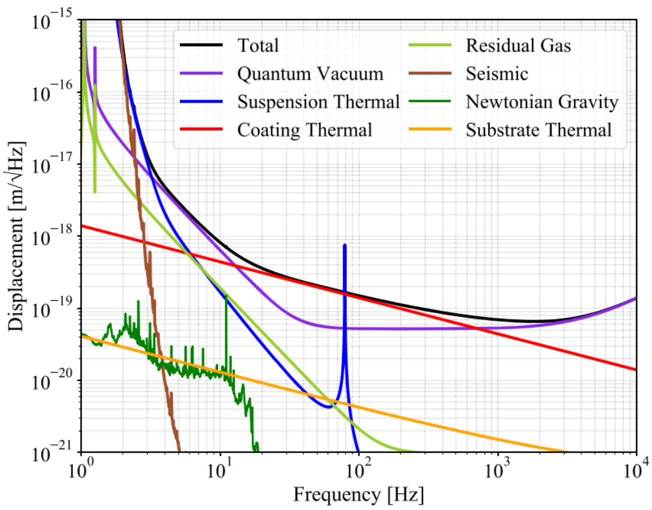


Figure 2.2: Amplitude spectral density of ETpathfinder's projected sensitivity [7]

The displacement sensitivity of ETpathfinder is close to that of the currently operating GW detectors. Its strain sensitivity, however, is too small to actually detect gravitational waves.

2.2 Calibration methods

Calibrating the absolute strain of a GW detector is essential because the interferometer does not directly measure strain. It measures the photodiode output voltage that must be translated into the relative displacement of the test masses. There are two approaches to calibrate the detector output: in-loop and out-of-loop calibration.

2.2.1 In-loop calibration

In-loop calibration of a detector relies on the structure of the control systems. To operate the interferometer, the cavities must be tightly tuned to resonance. The local control loops stabilise the optics by damping the suspension resonances and pre-align the mirrors. The global control loops then use these stabilised optics to *lock* the interferometer — keeping the arm cavities at resonance [11]. With in-loop calibration, the absolute differential arm length must be reconstructed from a network of coupled control paths, filters, and actuators, each including its own response and uncertainty.

Free-swinging Michelson A common method is the free-swinging Michelson method, where the interferometer is operated in a simple Michelson configuration with free test masses [10]. This allows the digital control signals to be mapped to the test mass displacement with a lower uncertainty. However, this method provides only a temporary calibration of the interferometer. That is why most detectors nowadays rely on out-of-loop calibration methods.

2.2.2 Out-of-loop calibration

Out-of-loop calibration provides an independent reference for the absolute displacement of the test masses by applying a known external force that bypasses most of the interferometer's control loops. This approach avoids the accumulated uncertainties of in-loop calibration, since the applied force does not rely on the same suspension stages, actuators, and filters used during operation.

Newtonian calibrator A Newtonian calibrator (Ncal) uses a rotating set of masses to generate a known, time-varying gravitational field near the test mass, thereby inducing a minuscule but measurable gravitational force as a reference [12]. This reference force depends on just the geometry, masses, and Newton's constant.

Photon calibrator A widely adopted method is the photon calibrator (Pcal), where an auxiliary power-modulated laser beam is directed onto the test mass. The resulting radiation force is proportional to the optical power, which can be measured with high accuracy. Because Pcals are a non-contact, well-understood, and traceable calibration method, they provide a robust calibration signal.

Recent research by Inoue et al. [13] has proposed a hybrid calibration method that combines an Ncal and Pcal system. In a nutshell, the Pcal is used as a precise actuator and the Ncal as an accurate absolute reference. The Ncal uses multiple rotor modulations to determine the exact distance between the Ncal and the test mass, thereby greatly reducing the systematic error. Then, the Pcal is calibrated against the Ncal by adjusting the power until the Ncal-induced displacement is cancelled. Thereby avoiding calibration of the photodiode. This method is expected to reach a lower calibration uncertainty than the individual methods.

2.3 Photon calibration principles

In this section, the fundamental principles of the photon calibrator are presented, which enable actuation of the test mass. It is shown how the output power of the Pcal beam is related to the induced displacement of the test mass.

2.3.1 Radiation pressure

The photon calibrator leverages the phenomenon of radiation pressure. Although a photon is a massless entity, it carries momentum p, which is defined by its energy E and the speed of light c.

$$p = \frac{E}{c} \tag{2.1}$$

The force F exerted by a photon is given by the time rate of change of its momentum. Hence, we differentiate (2.1) with respect to time to obtain

$$F = \frac{dp}{dt} = \frac{1}{c}\frac{dE}{dt} \tag{2.2}$$

The time rate of change of the energy is equal to the power P, while the speed of light is constant over time. For a specular reflection of light at normal incidence to a surface, twice its momentum is transferred. At an angle of incidence θ with respect to the normal, the exerted force can be expressed as

$$F = \frac{2\cos\theta}{c}P \tag{2.3}$$

Where P is the reflected power. Transmitted light does not transfer its momentum and therefore does not create a force. Absorbed light transfers its momentum once, creating a force in the direction of the incident beam.

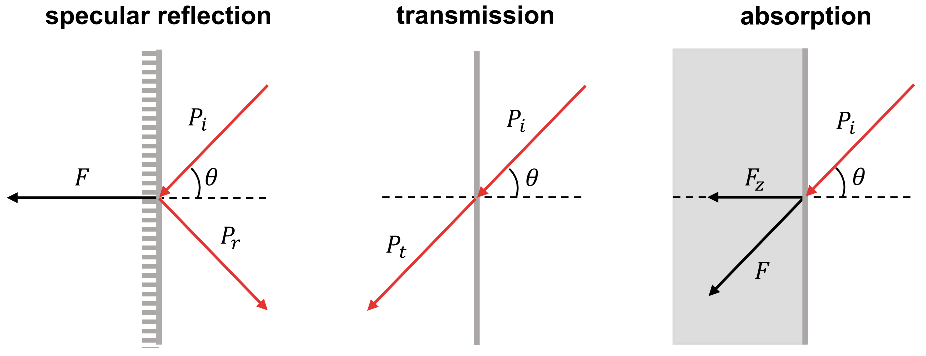


Figure 2.3: Radiation force F for specular reflection, transmission, and absorption

2.3.2 Force-to-displacement response

Given that the reflected laser power P is precisely measured, (2.3) will determine the force F applied to the test mass. However, to obtain the displacement z of the test mass over the modulation bandwidth, the force-to-displacement response $S(f) = z(f)/F(f)$ needs to be known such that,

$$z(f) = \frac{2\cos\theta}{c} S(f)P(f) \tag{2.4}$$

ETpathfinder's test masses have a mass of $m = 3.29$ kg and are suspended from the previous stage on silicon fibres with a length of $L = 0.4$ m [7]. Hence, the transfer function of force to displacement can be described by a simple pendulum with an eigenfrequency at

$$\omega_0 = \sqrt{\frac{g}{L}} = 5.0 \quad \text{rad/s} \quad \text{or} \quad f_0 = \frac{1}{2\pi}\omega_0 = 0.79 \quad \text{Hz} \tag{2.5}$$

with g, the gravitational acceleration. The test mass is excited at small angles so its translational response can be well-approximated by a *mass-spring-damper* system with mass m and damping coefficient d.

$$S(\omega) = \frac{1}{m(\omega_0^2 + \frac{d}{m}\omega i - \omega^2)} \tag{2.6}$$

with $\omega = 2\pi f$, the angular frequency. The damping is usually parametrised by the loss angle ϕ. Depending on the suspension wire material, the loss angle ranges from $\phi = 10^{-5}$ to 10^{-9}. The two are related to each other via $d = m\omega_0\phi$, which corresponds to a damping coefficient in the range of $d = 10^{-4}$ to 10^{-8}.

At frequencies far greater than the eigenfrequency $\omega \gg \omega_0$, the effects of resonance and damping are negligible, and the response can be written as a *free-body* response.

$$S(\omega) = -\frac{1}{m\omega^2} \quad \text{or} \quad S(f) = -\frac{1}{m(2\pi f)^2} \tag{2.7}$$

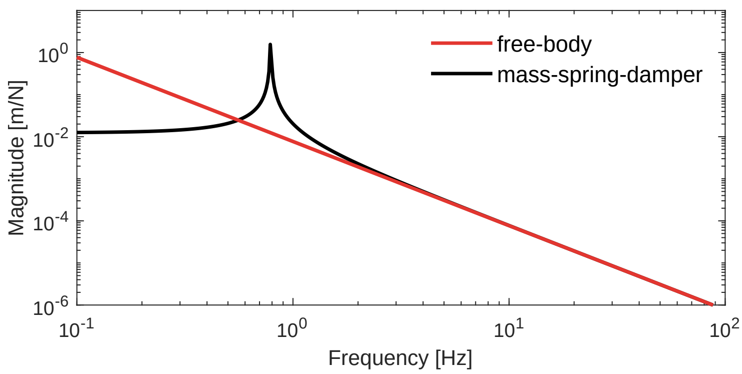


Figure 2.4: The estimated response compared to the actual response

The free-body approximation is used because the exact values of d and ω_0 are not well known, whereas the modulation frequency f and the mirror mass m are measured precisely.

By limiting the discrepancy between the actual response (2.6) and the free-body response (2.7), the calibration uncertainty can be minimised passively. It should be noted that, using measurement data, the parameters ω_0 and d could be estimated to reduce this uncertainty further. However, such analysis lies outside the scope of this thesis.

At the lower limit of the desired Pcal bandwidth, 10 Hz, the two responses differ by 0.62 %, whereas at 20 Hz this is reduced to 0.14 %. Altogether, we obtain the following expression for the induced mirror displacement as a function of the modulated laser power.

$$z(f) = -\frac{2\cos\theta}{c}\frac{1}{m(2\pi f)^2}P(f) \tag{2.8}$$

2.4 Conclusion

To detect the minuscule length changes induced by gravitational waves, a Michelson interferometer with Fabry-Perot cavities is used. Several methods exist to actuate the test mass with an out-of-loop reference to calibrate absolute arm-length variations. A photon calibrator leverages the effect of radiation pressure by reflecting a power-modulated laser beam off the test mass. At frequencies above 10 Hz, the force-to-displacement response is well approximated within 0.62% by a free-body response.

3 Optical lever as photon calibrator

The initial design of the ETpathfinder interferometer is currently under commissioning. In this chapter, a feasibility study is conducted to assess the implementation of a photon calibration system for the first phase of ETpathfinder. Specifically, an implementation in which the currently present optical lever sensor beam also serves as a Pcal beam.

Using numerical methods, the deformation of the test mass due to the localised Pcal force is quantified, and its effect on the calibration uncertainty is determined. The achievable signal-to-noise ratio is computed considering the source output power and the detector sensitivity. Finally, the dominant error sources in the Pcal system are identified, and their contributions to the calibration uncertainty are estimated.

3.1 Optical lever system

Optical levers are auxiliary laser beams that are reflected off the test mass and the suspension system, and are used to monitor and control the test mass orientation [11]. The reflected beams are measured with a position-sensitive detector. The front surface optical lever, indicated by the red beam in Figure 3.1, provides information on the pitch, yaw, and longitudinal position of the test mass. This information is used by the local control system to pre-stabilise the test masses during lock acquisition.

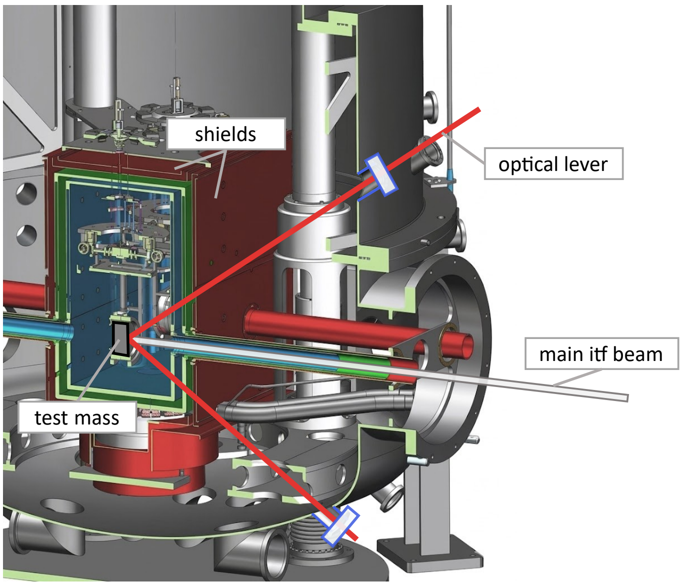


Figure 3.1: The optical lever beam path through the shield apertures

The optical lever transmitter and receiver boxes are mounted onto flanges on the outside of the vacuum tower. The beam path is in plane with the main interferometer beam and has an angle of incidence of $35°$ with respect to the mirror surface normal.

A potentially efficient implementation of a photon calibrator for ETpathfinder is to use the optical lever directly as a Pcal beam. The key advantage of this approach is that most of the required infrastructure is already present, whereas a separate photon calibration system would require additional equipment. By allowing the source and detectors to be shared by the two systems, this would be a cost-effective solution.

The radiation shields limit the physical and optical access to the test. This is why a number of small holes are drilled in the shields to allow the suspension wires and auxiliary beams to pass through. Another advantage of this Pcal implementation is that apertures are already present for the optical lever beam, avoiding the need for additional holes or modifications to the shields.

3.1.1 Superluminescent diode source

The optical levers use a fibre-coupled superluminescent diode (SLD) as their source, with a centre wavelength of 673 nm. This source produces a collimated beam, like a laser, but with low temporal coherence, similar to that of an LED. Similar to a laser, the SLD contains an electrically driven p–n junction and a waveguide. The difference, however, is that the SLD utilises spontaneous emission, followed by amplification, whereas a laser relies on stimulated emission with optical feedback [14].

In-fibre splitter

The SLD has an output power of 4.6 mW, and to reduce costs, the source is split into six beams using an in-fibre splitter [15]. Hence, a single SLD is used for the six optical levers in a single vacuum tower, resulting in output powers of 0.58 mW (12.5 %) and 0.86 mW (18.75 %) as shown in Figure 3.2.

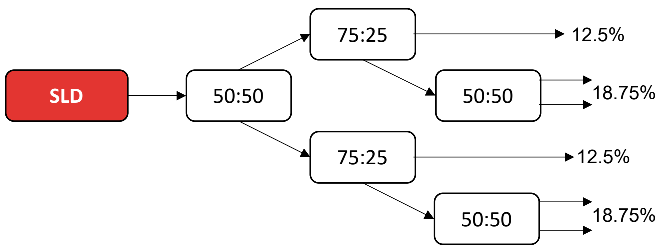


Figure 3.2: Schematic of the SLD in-fibre splitter tree [15]

The receiving position-sensitive detector uses a normalised position, so the absolute power on the detector should not influence the measured position. However, a different issue with modulating the source precludes the need for further investigation.

Common-arm actuation problem

Modulating the SLD source before it is split will cause all optical levers in a single vacuum tower to be modulated in phase. The front-surface optical levers will function as a Pcal beam, which should induce a relative change in the arm length between the two cavities. However, ETpathfinder is designed with parallel arm cavities (see Figure 1.2) with both ETMs in the same tower. Using the modulated optical levers as Pcal beams would extend and shorten both cavities by the exact same amount. This common-arm actuation produces no signal to the interferometer, so directly modulating the source is not an option.

It is decided that one of the front-surface optical levers will use a separate spare SLD source rather than the split fibre. Ultimately, the Pcal source can be replaced with a more suitable laser, but as a potentially cost-effective solution, the feasibility of the SLD source is analysed.

3.2 Test mass deformation

When a Pcal beam is reflected off the test mass, it causes some elastic deformation of the surface. Hild showed in [16] that this deformation introduces an uncertainty in the calibration of the test mass motion. To avoid this calibration error, the LIGO and KAGRA Pcal systems use two Pcal beams positioned away from the test mass centre [17, 18]. However, this is not possible at ETpathfinder because optical access is limited by the radiation shields.

A study is conducted on the effects of local and bulk deformation on the force-to-displacement response $S(f)$. *Local* deformation is a direct result of the induced force, whereas *bulk* deformation refers to the deformation of the surface through eigenmodes. More details on the steps taken and model parameters are provided in the Appendix section A.2.

3.2.1 Local deformation

To quantify the effects of local deformation, a COMSOL [19] simulation is performed to study the elastic deformation of the surface under a localised force. This study is based on the methods provided in [16] and [20]. A 2D axisymmetric model of the silicon test mass is constructed. To simulate the localised force from a Pcal beam, a Gaussian-distributed load is modelled at the centre of the mirror. It is assumed that the beam is reflected at normal incidence to the surface, such that the intensity profile $I_p(r)$ can be expressed as

$$I_p(r) = \frac{2P}{\pi w^2} e^{-2r^2/w^2} \tag{3.1}$$

with the optical power P, the beam waist w, and the radius r. In the case of perfect specular reflection of the light, the pressure profile $p(r)$ is given by

$$p(r) = \frac{2I_p(r)}{c} \tag{3.2}$$

To accurately resolve the local deformation at the load location ($r = 0$), a fine mesh is used in the vicinity of the load and a standard mesh size elsewhere (see Figure 3.3). The structural properties of silicon are dependent on its crystalline orientation [21]. However, for this study, the value for the Young's Modulus $E = 130$ GPa and Poisson's ratio $\nu = 0.28$ are taken from Table 3 in [22] and are assumed to be isotropic.

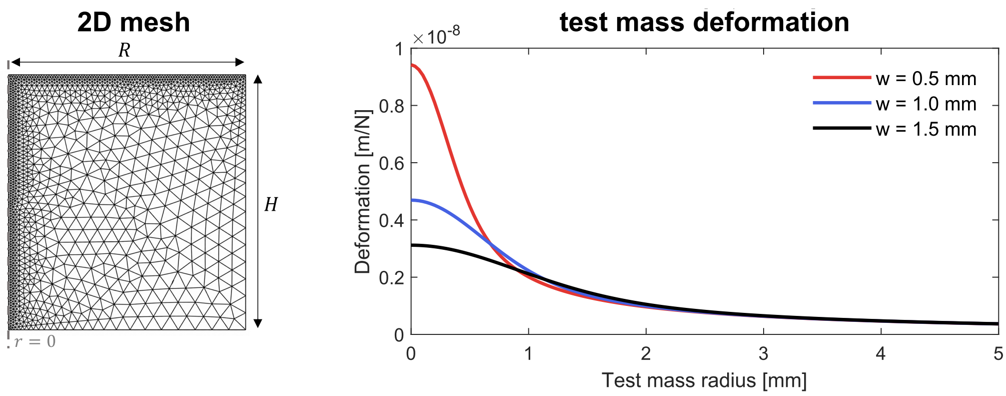


Figure 3.3: 2D mesh (left) and deformation of the test mass for various beam waists (right)

The optical lever beam waist radius w at the test mass surface is not exactly known. Hence, three values for the beam waist $w = \{0.5, 1.0, 1.5\}$ mm are considered and evaluated on their influence. On the right in Figure 3.3, the surface deformation per unit of applied force is plotted along the radial direction of the test mass surface.

The influence of this local deformation on the response $S(f)$ is quantified by the *effective displacement* D_{eff}, which denotes the displacement that is measured by the interferometer as a result of the deformed surface. This value is given by the weighted product of the main interferometer beam intensity, $I(r)$, and the corresponding displacement $D(r)$ [16]. For the radially symmetric case, this is expressed by the following integral.

$$D_{eff} = \int_0^R 2\pi\, r\, k_I\, I(r)\, D(r)\, dr. \tag{3.3}$$

with k_I, a normalization factor for $I(r)$ such that

$$\int_0^R 2\pi\, r\, k_I\, I(r)\, dr = 1. \tag{3.4}$$

These integrals are computed using numerical integration in Matlab [23]. For the three Pcal radii considered, the effective displacements are calculated using (3.3) and (3.4), and are displayed in the table below.

Table 3.1: The effective displacement for varying Pcal radii

Pcal waist radius [mm]	**Effective displacement [m/N]**
0.5	1.90×10^{-9}
1.0	1.80×10^{-9}
1.5	1.62×10^{-9}

3.2.2 Bulk deformation

An eigenfrequency study is conducted in COMSOL using a 3D model of the test mass to determine the effects of a modulated force on the bulk deformation. The details of this study are shown in section A.1. The first eigenmode at $f_1 = 17.8$ kHz is the *butterfly* mode, but this mode is not excited by a centred Pcal beam, and the effective displacement integrates to zero. The *drumhead* mode at $f_2 = 26.3$ kHz is excited by the Pcal beam.

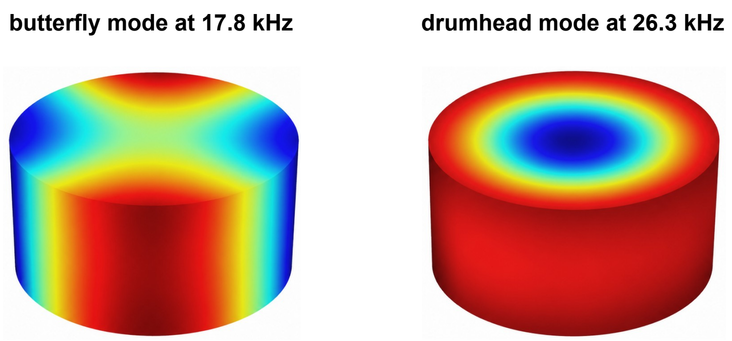


Figure 3.4: The butterfly and drumhead eigenmodes of the ETpathfinder test mass

In Figure 3.4, the colours a positive (red) or negative (blue) displacement of the surface. The drumhead mode is modelled as a resonance in the force-to-displacement response of the test mass, as explained further in the following section and shown in Figure 3.5.

3.2.3 Force-to-displacement response

The total response of the measured displacement as a result of an applied force can be written as the sum of the translational response $S_T(f)$ and the effective displacement by deformation $S_D(f)$. Where $S_D(f)$ represents the local and bulk deformation.

$$S(f) = S_T(f) + S_D(f) \tag{3.5}$$

The translational force-to-displacement response $S_T(f)$ of the test mass can be estimated as a free body in space as explained in subsection 2.3.2.

Deformation response

For frequencies far less than the drumhead eigenfrequency $f \ll f_2$, the deformation response $S_D(f)$ has a linear relation between force and displacement, represented by the effective displacement $D_{eff} = x/F = 1.90 \times 10^{-9}$ m/N. This displacement is in phase with the applied force. On the contrary, the free-body displacement $S_T(f)$ is 180° out of phase with the applied force, and decreases in magnitude with an f^{-2} slope.

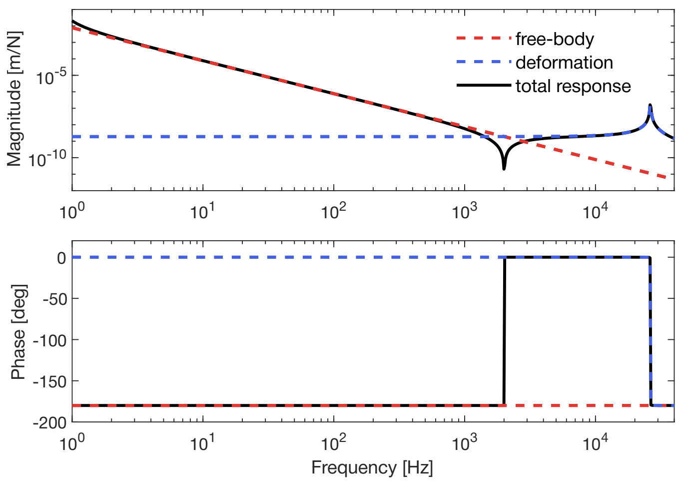


Figure 3.5: The total response of a Pcal beam force on the test mass with a notch at 2 kHz

The total response $S(f)$ is defined as the sum of the free-body and deformation response, as given by (3.5). Because these contributions have a 180° phase difference and equal magnitude, a notch occurs at $f = 2$ kHz. At this frequency, the interferometer cannot resolve the Pcal-induced displacement, resulting in a large calibration uncertainty at 2 kHz.

Measurement data can be used to determine a fit of the $S(f)$ transfer function, yielding a better estimate of the response, as is done at Virgo [24]. However, the exact position of the notch varies with the relative alignment of the interferometer and Pcal beams, and as the slopes in the vicinity of this notch are so steep, not all uncertainty can be eliminated. Conclusions about the effects of fitting the data can be drawn from experimental measurements, but this is beyond the scope of this thesis.

3.2.4 Response discrepancy

As argued in subsection 2.3.2, the free-body response can be taken as a first-order approximation for the relation between the Pcal force and test mass displacement. Deviation from this estimation can be treated as an upper bound of the calibration uncertainty. Hence, the fraction of the free-body response $S_T(f)$ over the expected total response $S(f)$ acts as a measure of the calibration uncertainty as a result of the test mass deformation.

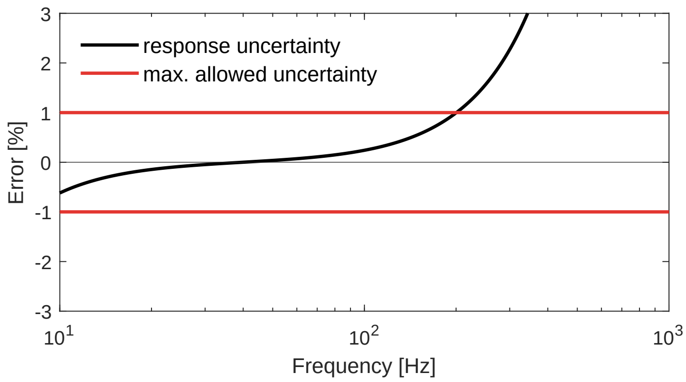


Figure 3.6: Calibration error due to test mass deformation

The discrepancy between $S_T(f)$ and $S(f)$ is shown in Figure 3.6, representing the calibration error over the desired bandwidth of 10 Hz to 1 kHz. The red line indicates the requirement for the total allowed error. The error at low frequencies arises because the free-body response underestimates the magnitude near the 0.79 Hz eigenfrequency. At higher frequencies, the discrepancy rapidly increases and exceeds the 1 % requirement around 200 Hz.

3.3 Test mass rotation

Another potential source of uncertainty is the apparent displacement measured due to the rotation of the test mass. When the Pcal beam centre of force $\vec{b}$ is not exactly centred on the test mass, it causes a rotation. If, in addition, the location of the interferometer beam $\vec{a}$ is not centred, this will result in a false measurement of displacement. Where $\vec{a}$ and $\vec{b}$ are vectors from the test mass centre to the beam spot position in the xy-plane, as visualised in Figure 3.7.

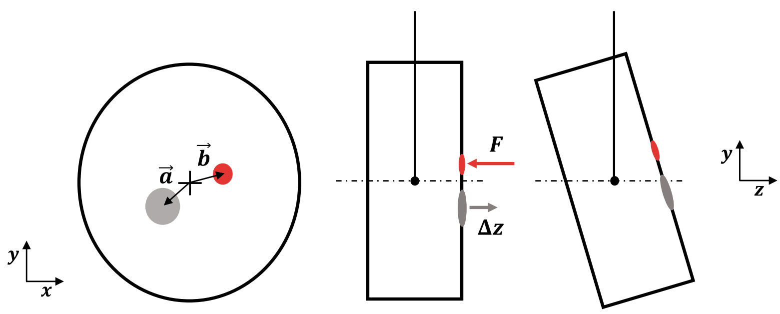


Figure 3.7: Apparent displacement due to offset Pcal and interferometer beams

The result of a force F on the apparent displacement Δz is captured by the rotational response $S_R(f)$ of the test mass. An off-centred force F creates a moment M around the test mass centre of mass. Consequently, the moment introduces a rotational acceleration $\ddot{\theta}$ around the radial axis.

$$-\vec{F} \times \vec{b} = M = I\ddot{\theta} \tag{3.6}$$

with the mass moment of inertia of a cylinder $I = \frac{1}{12}m(3R^2 + H^2)$. Performing a Laplace transform and rearranging terms gives

$$\theta(\omega) = -\frac{\vec{F} \times \vec{b}}{I\omega^2} \tag{3.7}$$

Ultimately, the relation between the angle θ and the displacement Δz is taken to arrive at the expression for the force F to displacement Δz response due to rotation.

$$S_R(\omega) = -\frac{\vec{a} \cdot \vec{b}}{I\omega^2} \tag{3.8}$$

It can be concluded from (3.8) that either having the Pcal beam or the interferometer beam perfectly centred will not introduce an apparent displacement. At this moment, it is unclear what the stability of the interferometer beam position $\vec{a}$ will be at ETpathfinder. Still, the calibration error will be evaluated for several cases to establish a requirement on the relative stability of the Pcal and main interferometer beams.

3.3.1 Response discrepancy

To arrive at the resulting calibration error, the discrepancy between the rotational response (3.8) and free-body response (2.7) is taken as an upper bound value. Similar to the approach in subsection 3.2.4, but now the error is independent of frequency.

$$\frac{S_R(\omega)}{S_T(\omega)} = \frac{m\,(\vec{a} \cdot \vec{b})}{I} \tag{3.9}$$

In Figure 3.8, the error is shown for various offsets of the interferometer and Pcal centre of force, where the vectors $\vec{a}$ and $\vec{b}$ are assumed to be parallel.

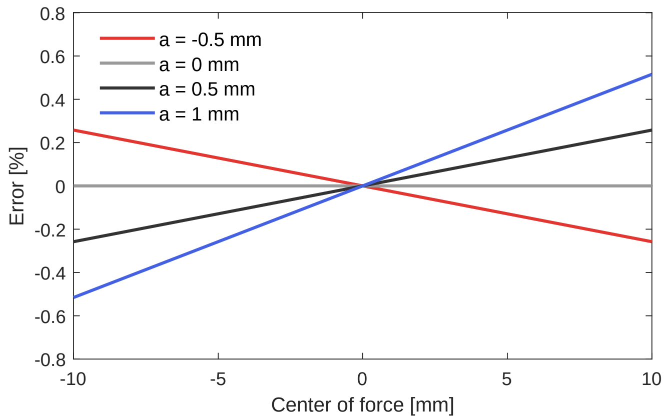


Figure 3.8: Calibration error due to rotation as a function of b for various a

At Virgo, the interferometer offset is controlled to be better than $\Delta a \pm 0.5$ mm [24]. It is expected that ETpathfinder will achieve a similar level of stability; hence, for this analysis, the same variation is considered. Assuming the shield apertures are aligned with the test mass centre, and the Pcal beam passes through the 20-mm diameter shield apertures, the maximum variation of the centre of force is taken as $\Delta b \pm 10$ mm. Altogether, the estimated uncertainty due to rotation of the test mass is $\sigma_{rot} = 0.26$ %.

3.4 Signal-to-noise ratio

Given the calibration uncertainty requirement, a minimum signal-to-noise ratio (SNR) is imposed on the Pcal-induced displacement signal. As the relative uncertainty is given by the inverse of the SNR, a calibration uncertainty below 1 % requires an SNR > 100 across the bandwidth from 10 Hz to 1 kHz. The SNR is defined as

$$SNR = \frac{\text{signal amplitude}}{\text{noise amplitude}} = \frac{z(f)}{\Delta L(f)/\sqrt{T}} \tag{3.10}$$

with $z(f)$ the Pcal-induced displacement in m, $\Delta L(f)$ the amplitude spectral density of the interferometer noise in m/$\sqrt{\text{Hz}}$, and T the measurement time in s. The signal amplitude can be written as a function of the laser power P and the force-to-displacement response $S(f)$. The projected displacement sensitivity $\Delta L(f)$ for ETpathfinder is shown in Figure 2.2.

In order to get an estimate of the SNR, an expression of the induced displacement $z(f)$ needs to be derived. This depends on the reflectivity of the source on the test mass coating.

3.4.1 Reflectivity of the SLD

The superluminescent diode (SLD) source has a centre wavelength of 673 nm. Its optical spectrum, taken from the specification sheet [25], is shown in red in Figure 3.9. The left y-axis shows the simulated spectral reflectivity of the test mass reflective coating, supplied by the coating manufacturer. The reflectivity is shown for a beam at an angle of incidence (AOI) of 35°, for both s-polarisation (solid line) and p-polarisation (dotted line).

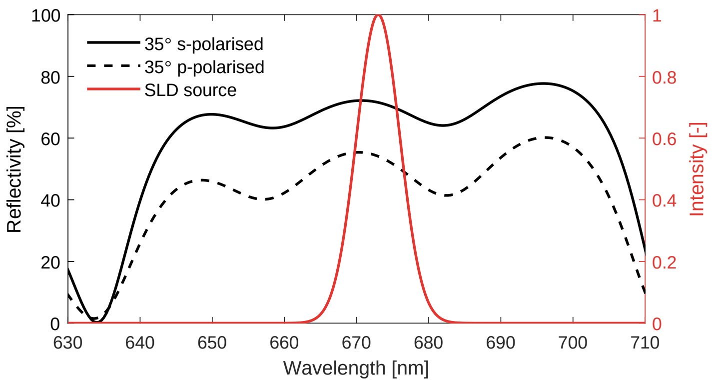


Figure 3.9: Spectral reflectivity of the test mass coating at an AOI of 35° (black) and the normalised intensity of the SLD source (red)

By taking the weighted average of the reflectivity, the total reflectance is $\rho = 0.71$ for s-polarised, and $\rho = 0.53$ for p-polarised light. The SLD source produces a linearly polarised beam, so it is assumed that 71 % of the incident power is reflected. At this wavelength, the remaining 29 % of the SLD power is absorbed by the silicon test mass. This results in a radiation-pressure force in the direction of the incident beam (see Figure 2.3), while the absorbed photons transfer their momentum once to the test mass. The z-component of the radiation force through absorption is given by

$$F_z = (1-\rho)\frac{\cos\theta}{c}P \tag{3.11}$$

Compared to the case of total specular reflection (2.3), the induced force and SNR are reduced. Moreover, the reflectivity ρ is subject to uncertainty, which adds to the total calibration uncertainty budget as will be explored further in section 3.5.

3.4.2 Achievable SNR

The SLD has an output power of $P = 4.6$ mW, and the angle of incidence of the optical lever is fixed to $\theta = 35°$. The integration time T is set to a maximum of 60 s, but should ideally be less than or equal to 10 s. This is because the calibration of GW detectors must be performed continuously during operation of the interferometer across multiple frequencies to account for unknown variations that might affect the absolute strain amplitude.

$$SNR = \frac{P\,(1+\rho)\cos\theta\, S(f)\,\sqrt{T}}{\Delta L(f)\,c} \tag{3.12}$$

The SNR is calculated for various frequencies as shown in Table 3.2, with each having a proper integration time in order to reach the SNR > 100 requirement.

Table 3.2: Achievable SNR at 4.6 mW at various frequencies f and integration times T

Frequency [Hz]	SNR with T=10 s	SNR with T=60 s
50	1175	2880
100	368	902
200	109	268
350	41	100

The signal-to-noise ratio drops quickly for increasing frequency because of the f^{-2} slope in the response $S(f)$. By extending the measurement time to 60 s, the SNR requirement can be met up to 350 Hz. Since the SNR scales with the square root of integration time, further increasing T does not significantly affect the SNR.

Ways of achieving an SNR of 100 over the full bandwidth up to 1 kHz modulation will be investigated further in chapter 4. For now, it is concluded that the SLD satisfies the SNR requirement up to 350 Hz under the assumption of an ideal modulation signal. In practice, this signal is non-ideal, which will be investigated in the next section for the SLD.

3.4.3 Power modulation

Modulation of the Pcal laser power at current GW detectors is achieved using an acousto-optic modulator (AOM) [17], or by modulating the pump diode current of the laser [18, 24]. Ideally, the output power should follow a sinusoidal waveform, such that the magnitude of the modulation frequency exceeds that of the harmonics and the laser power noise.

SLD modulation experiment

It is experimentally tested whether the SLD source module can modulate the output power to the desired waveform. The experimental setup is shown in section B.3. A waveform generator is used to input a 10 Hz sine wave ranging from 2 to 4 V into the source module.

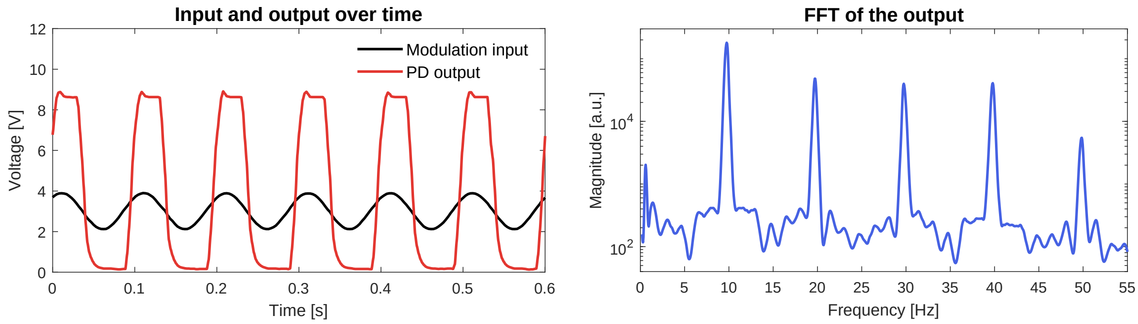


Figure 3.10: Time series (left) and frequency plot (right) of the power modulation at 10 Hz

It can be seen in Figure 3.10 that the output power (red) does not track the requested waveform (black). This is because the module housing the SLD uses an optocoupler to isolate the electronic circuit from the source. This causes on/off switching of the output power, resulting in the harmonics not being sufficiently suppressed. The SNR of the modulated frequency (10 Hz) relative to the harmonics (20 Hz, 30 Hz, ...) should ideally be greater than 100, but it currently appears to be less than 10. A new method for modulating the power is required.

Possible solutions

Analogue current modulation of the SLD source could be a solution to this problem if implemented cheaply using custom electronics. Alternatively, an acousto-optic modulator (AOM) could be placed in the transmitter beam path to modulate the power, although it is relatively expensive. In that case, it might be better to invest in a new source with an analogue driver. Further research on this is required before a Pcal system is implemented at ETpathfinder.

3.5 Static error sources

In addition to the response errors discussed in section 3.2 and section 3.3, other sources contribute to the total calibration uncertainty. In principle, these static error sources can be derived directly from the equation that relates the modulated laser power to the test mass displacement.

$$z(f) = -\frac{(1+\rho)\cos\theta}{c}\frac{1}{m(2\pi f)^2}P(f) \tag{3.13}$$

3.5.1 Angle of incidence θ

The angle of incidence θ of the optical lever beams is constrained by the viewport location and the shield apertures. From the ETpathfinder CAD drawings, the angle is estimated to be 35°, corresponding to 0.61 radian. The stability of the beam angle determines the uncertainty, but this is not known exactly.

For now, a worst-case estimate of the stability is made by taking the maximum possible variation of the beam angle $\theta \pm \Delta\theta$ for which the beam still passes through the shield apertures. At a distance of 1.5 m from the viewport, the beam must pass through a 20 mm-diameter aperture. Resulting in a maximum angle variation of $\Delta\theta = 6.7$ mrad.

Taking into account the cosine in (3.13), this results in a worst-case uncertainty on the angle of incidence of $\sigma_\theta = 0.47$ %.

3.5.2 Mirror mass m

The mirror mass m of ETpathfinder's test mass is 3.29 kg. However, a calibrated measurement of its exact mass is required to make a definitive statement about the uncertainty. Test masses of the Virgo [24] and LIGO [17] detectors have a measurement uncertainty of 0.17 % and 0.05 %, respectively. Assuming a comparable measurement accuracy is available for ETpathfinder, a conservative uncertainty of $\sigma_m = 0.2$ % is taken.

3.5.3 Reflectance ρ

According to the coating specifications for the ETpathfinder test masses, a maximum difference in reflection of 0.1 % between the two ETM substrates is allowed. However, this does not imply that the reflectance is within 0.1 % of the theoretical specification, which would require an absolute reflectance measurement at 673 nm.

The uncertainty on ρ may ultimately prove to be larger, but for now the uncertainty on the reflectance is assumed to be within specification, with $\sigma_\rho = 0.1$ %.

3.5.4 Optical efficiency η

The measured incident and reflected power are given by $\tilde{P}_i$ and $\tilde{P}_r$, respectively. These are measured at the transmitter and receiver modules outside the vacuum tower. In (3.13), the power $P(f)$ is in fact the laser power P_i right before the beam is incident on the test mass, and the actual laser power right after reflection is given by $P_r = \rho P_i$.

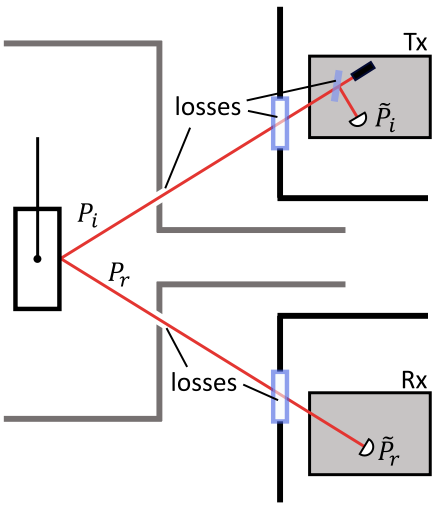


Figure 3.11: Sources of optical power loss from transmitter (Tx) to receiver (Rx)

Assuming the reflectance and power measurements are ideal, the in-situ measured efficiency η will be lower than the coating reflectance ρ due to optical power loss from passing through the viewports or shield apertures.

$$\eta = \frac{\tilde{P}_r}{\tilde{P}_i} \leq \frac{P_r}{P_i} = \rho \tag{3.14}$$

It is particularly difficult to determine the actual values of P_i or P_r. For instance, if the measured value of $\tilde{P}_r$ decreases by 1 %, it is unclear whether this reduction occurred before or after reflection. Therefore, this is treated as an uncertainty in the calibration. To minimise this uncertainty, current Pcal systems aim for a high optical efficiency, $\eta > 99.5$ %, from transmitter to receiver [17]. This is not possible due to the reflectance of the SLD source, so the aim is to minimise the discrepancy between η and ρ.

This uncertainty can only really be determined by measurements. Under ideal conditions, it is limited by the viewport AR coatings optimised at 673 nm, with a reflectance of 0.25 % each. The estimated value of the power P is then given by

$$P = P_i \approx \tilde{P}_i(1 - \frac{\rho - \eta}{2}) \tag{3.15}$$

The actual value lies between the upper and lower limit, treated as a rectangular distribution, resulting in $\sigma_\eta = (\rho - \eta)/2\sqrt{3} = 0.14$ %.

3.5.5 Photodiode calibration

A significant contributor to the calibration uncertainty of current Pcal systems is the measurement uncertainty of the power sensors. Current detectors use an InGaAs photodiode coupled to an integrating sphere to measure the Pcal power. The measurement uncertainty of these sensors translates directly into an uncertainty on the induced displacement.

NIST tractability

Calibration of the power sensors at the Pcal systems of the LVK detectors is done using the so-called *Gold Standard* (GS) power sensor, which is calibrated yearly at the National Institute of Standards and Technology (NIST) in the US. Each GW detector receives a *Working Standard* (WS) calibrated against the GS, which in turn is used to calibrate the transmitter and receiver power sensors of the Pcal system.

This traceable calibration method results in an uncertainty of $\sigma_{cal} = 0.6$ % [26]. The photodiodes are calibrated at 1047 nm, as this is the wavelength used by the current Pcal systems. Unfortunately, ETpathfinder and ET cannot use the same calibration infrastructure because they require a different Pcal wavelength, as will be explored further in chapter 4.

Hybrid Ncal calibration

Recent research on hybrid calibration has shown promising results for calibrating the Pcal photodiodes [13]. This method removes the need for traceable calibration of the power sensors by using an additional Ncal system to calibrate the Pcal power, as explained in subsection 2.2.2. It is expected that such a hybrid calibration method can reduce the calibration uncertainty to $\sigma_{cal} = 0.17$ %.

3.6 Calibration uncertainty budget

All sources of uncertainty are assumed to be uncorrelated and are therefore added in quadrature according to the following expression:

$$\sigma_{tot} = \left(\sum_{i=1}^{n} {\sigma_i}^2 \right)^{\frac{1}{2}} \tag{3.16}$$

with n the number of sources, as presented in Table 3.3.

Table 3.3: Frequency-independent calibration uncertainty budget for the optical lever Pcal

Parameter	**1σ uncertainty**
Angle of incidence [a]	0.47 %
Mirror mass	0.20 %
Reflectance	0.10 %
Optical efficiency [b]	0.14 %
Test mass rotation [a]	0.26 %
Photodiode calibration	0.17 %
Total	0.62 %

[a] Worst-case uncertainty based on geometrical constraints
[b] In practice, in-situ measurements of the power determine the uncertainty

The dominant error sources in Table 3.3 are the angle of incidence and test mass rotation. In this analysis, the worst-case angle of incidence and a conservative estimate of the rotation have been considered. Both these uncertainties can be minimised by ensuring a stable beam angle and spot position.

In addition to the static error sources, the frequency-dependent response error due to the test mass deformation is significant at higher frequencies. This uncertainty is added to the total budget in quadrature and plotted in Figure 3.12.

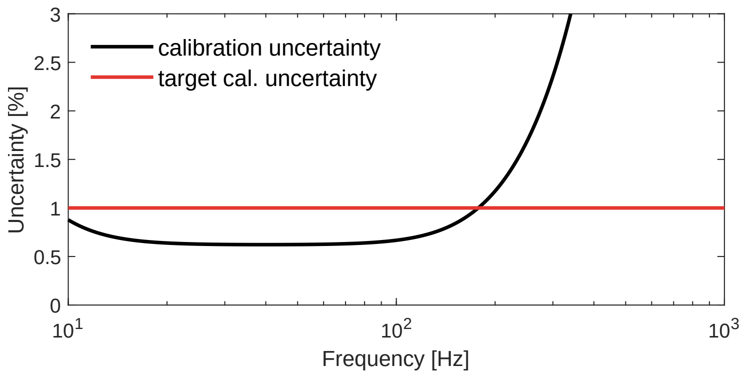


Figure 3.12: Frequency-dependent calibration uncertainty budget for the optical lever Pcal

Above 180 Hz, the requirement on the total calibration uncertainty of 1 % is exceeded. Near 10 Hz the uncertainty increases slightly due to the resonance of the pendulum mode, but remains below the requirement.

3.7 Conclusion

In this chapter, a feasibility study has been conducted on the implementation of a photon calibrator for ETpathfinder that uses the optical lever sensor to displace the test mass. The concept has been shown to be viable to a limited extent, although it requires several changes and additions to the system.

To avoid common-arm actuation, a spare SLD source must be used for one of the optical levers. Modulation of the output power is not possible with the current source module, and further research is necessary to identify a suitable alternative, e.g. an AOM or an analogue current driver. The SNR requirement is met up to 350 Hz due to limited output power and the SLD's partial reflectivity. Local deformation of the test mass centre leads to an overly large calibration uncertainty at frequencies above 180 Hz. Overall, this concept can be used to reliably calibrate the test mass displacement at lower frequencies.

4 Photon calibrator concepts

In chapter 3, it was shown that a power-modulated optical lever beam is not the ideal photon calibrator. The local deformation of the test mass centre results in a significant response error at all frequencies above 200 Hz. In addition, the SLD source can only produce sufficient SNR up to 350 Hz, and its suboptimal reflectivity introduces additional systematic errors.

In this chapter, methods to reduce the deformation-induced error are explored, and several photon calibrator concepts are considered. It is analysed which source characteristics are required to achieve the desired SNR and a sub-percent calibration uncertainty.

4.1 Reduced effective displacement

To minimise the dominant source of uncertainty at high modulation frequencies, the local deformation of the test mass near the interferometer beam needs to be reduced. In particular, the effective displacement due to this deformation should be less than 1% of the rigid body motion of the test mass. It has been concluded that increasing the beam diameter or using two offset beams is a possible solution.

4.1.1 Increasing beam radius

The 2D axisymmetric model from the effective deformation analysis is used in this section to explore the effects of increasing the beam waist radius w. Specifically, whether it is possible to reach a response error <1 % over the full frequency range of 10 Hz to 1 kHz.

A centred Gaussian load is considered. The deformation profiles for various beam radii w are plotted Figure 4.1, with D_{eff} the effective displacement. On the right, the resulting response uncertainty is shown, defined as $S(f)/S_T(f)$.

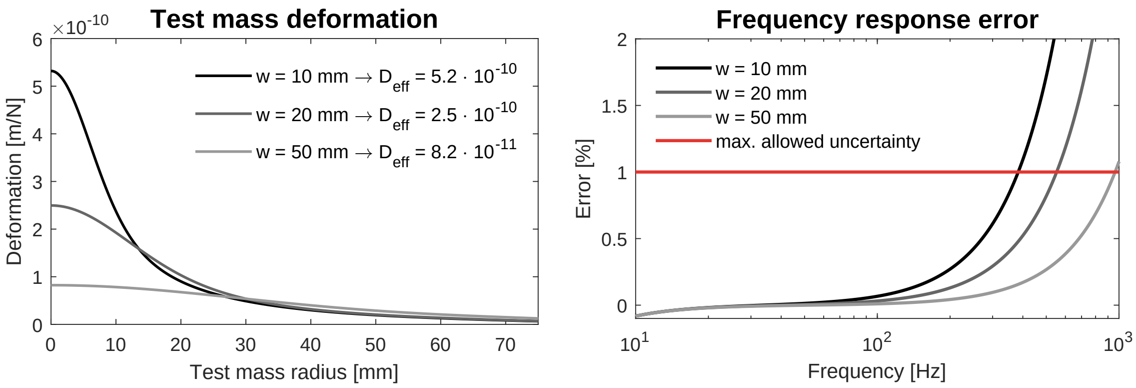


Figure 4.1: Deformation of the test mass for various beam radii w (left) and the resulting uncertainty on the response (right)

These results show that a minimum beam radius of 50 mm is required to meet the requirement. However, the contribution of other error sources will likely cause the total uncertainty to exceed 1 % near 1 kHz. Nonetheless, a widened beam could be a potential solution.

4.1.2 Splitting beams

To model the deformation due to split beams, a separate 3D model is created in COMSOL and validated against the 2D model for a centred beam. More details on the model parameters and approach are presented in section A.3. The split beams are modelled equidistant from the test mass centre with an offset d, as shown in Figure 4.2.

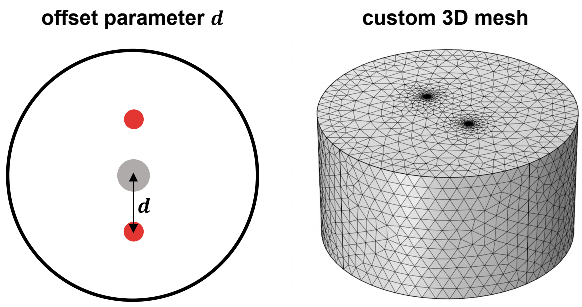


Figure 4.2: Visualisation of the beam offset d (left) and the 3D mesh for simulation (right)

The beams have a waist radius of 0.5 mm and a force of 0.5 N, making the results comparable to a single beam of 1 N, and allowing the displacement to be expressed in m/N. The model outputs a displacement field Δz of the surface in the z-direction on an irregular mesh. In Matlab, these data points are interpolated onto a grid with uniform spacing Δx and Δy. On this grid, the normalised intensity profile $I(x, y)$ of the interferometer beam is computed.

The continuous area integral, representing the effective displacement D_{eff}, is approximated by the double sum over all grid points given by

$$D_{eff} = k_I \sum_{j=1}^{n_x} \sum_{i=1}^{n_y} I_{ij} W_{ij} \Delta x \Delta y \tag{4.1}$$

with k_I a normalisation factor.

The effective displacement is calculated for various offsets d, and the response discrepancy as a result of those is plotted in Figure 4.3.

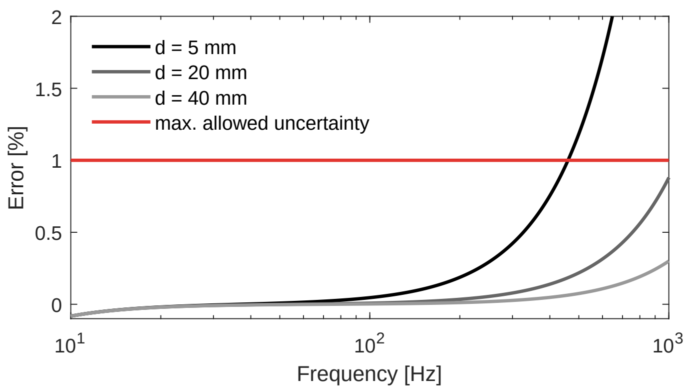


Figure 4.3: Uncertainty on the response for various split beam offsets

Taking into account the additional uncertainty from other sources, the beams must be offset by at least $d = 35$ mm from the centre, at which the resulting error is ~1 % at 1 kHz.

Optimal spot positions

The ETpathfinder interferometer is expected to be sensitive to frequencies up to approximately 10 kHz [7]. It is preferred, but not required, to achieve a <1 % calibration uncertainty at frequencies in the kHz range. Therefore, a brief study is presented on the optimal offset d of the two split Pcal beams to mitigate the errors caused by bulk deformation.

The Gaussian loads are modelled similarly to real Pcal beams by modulating them at a specified frequency. Using the surface integration tool in COMSOL, the effective displacement D_{eff} is calculated directly from the displacement field. Additionally, the centre-of-mass (COM) motion of the complete test mass is determined using volumetric integration. In Figure 4.4, the difference between D_{eff} and the COM displacement is plotted for various frequencies of the modulated loads. More details on this study are provided in section A.4.

Pcal systems, such as those at LIGO or KAGRA, have positioned their beams on the nodal circle of the drumhead mode, where the displacement is zero. At this location, the split beams do excite the lower-frequency butterfly mode, but in this mode, the effective displacement integrates to zero.

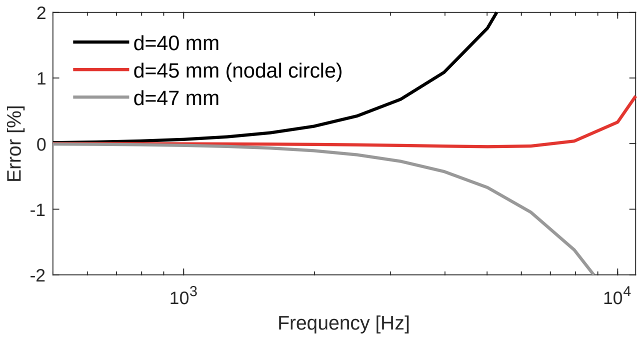


Figure 4.4: Uncertainty on the response due to bulk deformation of the drumhead eigenmode for various beam offsets

Excitation of the drumhead mode is strongly reduced for split beams located on the nodal circle. To minimise the response error due to bulk deformation at frequencies above 1 kHz, the optimal position for split beams is found to be at $d = 45$ mm.

Force imbalance due to mirror curvature

When the beams are split vertically and reflect off the test mass at an angle, the angle of incidence with respect to the surface differs slightly between the beams. For beams split by a distance of $d = 40$ mm at an angle of $\theta = 35°$, the effective radiation force changes by less than 0.001 %. The centre of force shifts down by 110 μm, resulting in a less than 0.02% error due to rotation of the test mass. Detailed calculations are presented in section B.1

4.2 Beam path concepts

Based on the results of the previous section, several concepts are explored for the Pcal beam path. The first three concepts aim to circumvent the limited optical access imposed by the current shield apertures, whereas the latter concepts require a redesign of the shields.

4.2.1 Main beam pipe entrance

At the KAGRA gravitational wave detector in Japan, the Pcal beams must also avoid interference with the cryogenic shields. This is accomplished by mounting the transmitter and receiver on a separate vacuum tower located 36 m in front of the ETM tower [18]. The Pcal beams are directed parallel to the interferometer beam, enter the radiation shields through the main aperture and reflect perpendicular to the test mass surface. As shown in Figure 4.5, a similar approach could be used at ETpathfinder

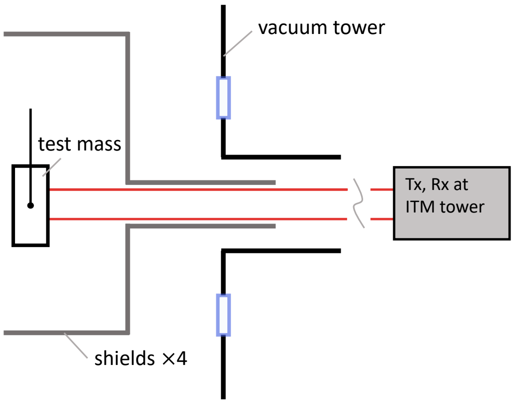


Figure 4.5: Concept of the beams entering through the main beam pipe

The main beam passes through a 50 mm-diameter aperture at the inner radiation shield, implying that the Pcal beam can have a maximum offset of $d = 25$ mm. This is below the minimal required offset determined in the previous section. Therefore, this concept is not considered a suitable solution.

4.2.2 Diverging beam

This concept utilises a single beam that passes through the existing apertures in the radiation shields. Inside the cryo chamber, the beam is diverged to increase its diameter at the test mass surface, which can be achieved with a concave lens or mirror. The reflected beam must be converged and collimated by a series of lenses or mirrors to pass through the aperture to the receiver module.

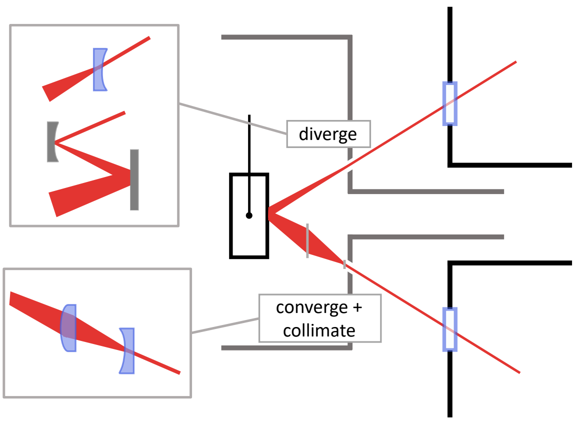


Figure 4.6: Concept of a diverging beam inside the radiation shields

Benefits By diverging the beam after it has passed through the apertures, no modifications to the radiation shields are required. Diverging and converging the beam can be done with relatively few optical components.

Limitations The beam waist radius must be at least 50 mm at the test mass surface, as concluded in the previous section. Converging and collimating the reflected beam will require an even larger aperture of a lens or mirror. Placing the optics within the cryochamber will significantly increase the complexity of mounting and positioning, since the components must be cooled to and maintained at temperatures near 15 K.

This concept is not considered a viable option due to its limited effectiveness in minimising uncertainty and the complexity of positioning the optics in the cryogenic system.

4.2.3 Split beams

The second concept splits the incident beam into two parallel beams inside the cryo chamber. This can be achieved using a beam splitter and additional folding mirrors to direct the beams to their desired positions on the test mass. To direct the beams back through the shield apertures, they must be recombined closely together, which can be achieved with mirrors, as shown in Figure 4.7.

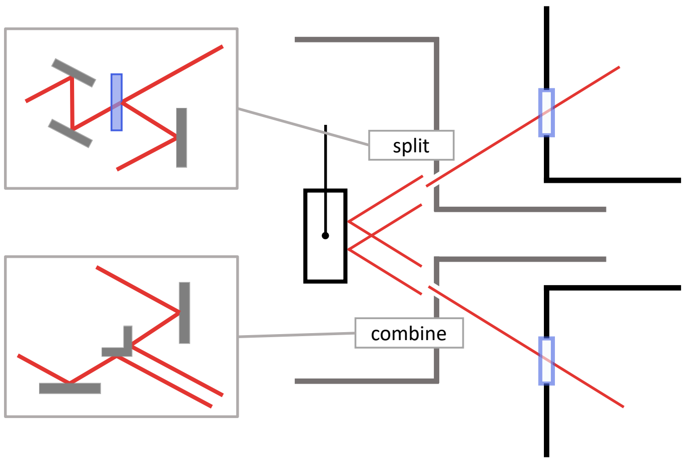


Figure 4.7: Concept of split beams inside the radiation shields

Benefits Split beams reduce the response error much more effectively compared to a widened beam waist. By locating the optics inside the cryo chamber, no changes to the current radiation shields are necessary.

Limitations A relatively large number of optical components is needed to split and redirect the beams into their desired parallel paths. The same holds for recombining the reflected beams and precisely directing them through the bottom shield apertures. Having such complexity in a 15 K vacuum environment creates many challenges. Moreover, for a prototyping facility such as ETpathfinder, it is advantageous to have access to optical components to test various system configurations. That is not possible with this concept.

This concept is not considered to be a feasible solution due to the complexity and impracticality of positioning the optics within the cryogenic system.

4.2.4 Split beams with cryogenic windows

The third concept also uses a two-beam configuration, but in this case, the beams are split and made parallel before entering the vacuum tower. Windows are placed within the shield to allow the beams to enter and exit the cryo chamber without compromising cryogenic system cooling the test mass to 15 K. This may consist of four windows, one per shield, but the optimal configuration remains to be investigated. The beam angle and line of sight are insensitive to any rotation or translation of a thin window, making the positioning of the windows non-critical.

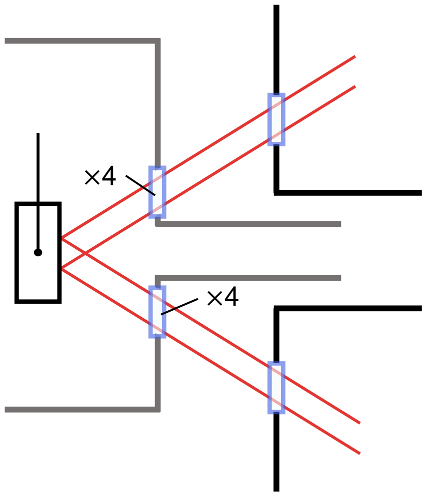


Figure 4.8: Concept of split beams with windows in the radiation shields

Benefits With this concept, there are no additional optical elements within the cryo chamber, enabling simpler access to these components for alignment and potential replacements. Moreover, the windows allow monitoring of the test mass with a camera, which can assist in alignment of the Pcal spot positions.

Limitations The radiation shields need to be redesigned and manufactured to accommodate the windows. Such a change will be made in a later stage of ETpathfinder.

4.2.5 More cryogenic window concepts

Additional concepts are considered that also rely on adding windows to the shields. These are shown in Figure 4.9 and allow the number of required windows to be reduced.

The first concept (on the left in Figure 4.9) considers a collocated transmitter-receiver module with an offset incident beam. After reflection from the test mass, a retroreflector or prism folds the beam back onto the test mass with the same offset, effectively mimicking the split beam configuration.

Benefits Using a retroreflector ensures that the incoming and outgoing beams are parallel. Given that the test mass and retroreflector are well aligned, an offset of the incident spot position will result in an equal and opposite offset of the reflected beam. Thus, the position of the centre of force is insensitive to misalignment of the spot position.

Limitations A drawback of this concept is that the retroreflector is difficult to access, as it is located within the cryogenic system. If the optical efficiency of the retroreflector is low or degrades over time (e.g., due to ice formation on the surface [27]), the reflected power may be significantly lower than the incident power, resulting in an offset centre of force. The mounting, positioning, and cooling of this component also introduce additional challenges.

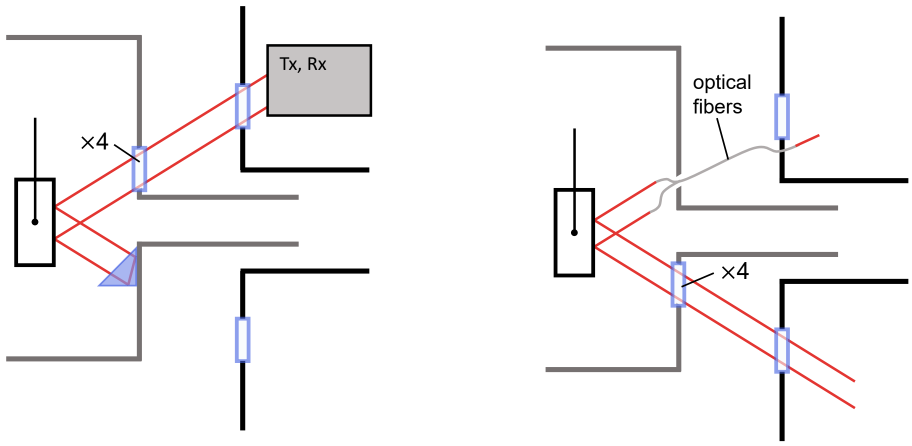


Figure 4.9: Concepts of using a retroreflector to fold the beam (left) and using optical fibres to move the input beams into the cryo chamber (right)

The second concept (on the right in Figure 4.9) uses optical fibres to exclude the need for windows for the transmitter beams. In principle, the reflected beams can also be coupled into optical fibres, but this is difficult to realise in practice.

Benefits Using optical fibres, the viewport in the vacuum tower need not be used, and the fibres can enter through the aperture before being split into the two-beam configuration.

Limitations A drawback of optical fibres is that the optical power of the beam exiting the fibre is not measured. Even if the efficiency is measured beforehand, low temperatures will negatively affect optical efficiency, but the magnitude of this effect remains uncertain.

Overall, these concepts have some favourable benefits and could reduce the number of cryogenic windows needed. Still, both concepts are rejected because they introduce additional error sources with large calibration uncertainties.

4.2.6 Motivation on cryogenic windows

The split-beam path with cryogenic windows, as described in subsection 4.2.4, is selected as the concept for the ETpathfinder's photon calibration system.

Although replacing the windows with two small apertures could be a simpler and more effective solution to this problem, this option is deliberately not considered. ETpathfinder is a prototyping facility intended for the research and testing of new technologies, and the concept of cryogenic windows is expected to be valuable in many other aspects of ETpathfinder and ultimately the Einstein Telescope. Therefore, this photon calibrator concept has been selected to investigate the design requirements, limitations, and possible configurations of cryogenic windows, serving as a stepping stone toward other applications in which cryogenic windows could provide a solution.

4.3 Laser source

The optical lever SLD source is not ideal for use as a Pcal beam. The partial reflection of 673 nm light introduces additional calibration uncertainty, and the 4.6 mW output power limits the SNR at high modulation frequencies. In this section, the optimal wavelength and laser power are determined to satisfy both requirements.

4.3.1 Optimal wavelength

The highly reflective (HR) coatings on the ETMs are specifically designed for the 1550 nm interferometer wavelength at normal incidence. The coating reflectivity depends on the incident angle, wavelength, and polarisation of the light.

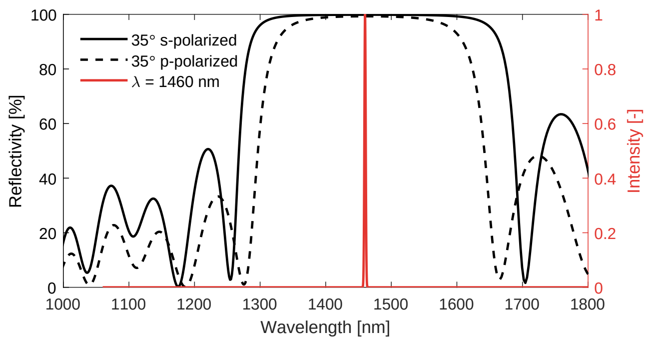


Figure 4.10: Spectral reflectivity of the test mass HR coating at an AOI of 35°

The maximum reflectance $\rho = 99.86$ % is achieved for 1460 nm s-polarised light. However, any wavelength between 1350 - 1580 nm will yield a reflectance of $\rho > 99.5$ %.

Viewport AR coatings

When a new Pcal laser wavelength of 1460 nm is used, the anti-reflective (AR) coatings on the vacuum tower viewports must be replaced. Currently, these are designed to optimise transmission of the 673 nm optical lever beams, so a new AR coating is necessary to maximise the optical efficiency from the transmitter to the receiver module. The same holds for the cryogenic windows.

4.3.2 Required laser power

The equation for the signal-to-noise ratio has been derived in section 3.4. With the Pcal wavelength at the optimal reflectivity of the test mass, the displacement amplitude $z(f)$ equals (2.8). Resulting in the following expression for the SNR.

$$SNR = \frac{z(f)}{\Delta L(f)/\sqrt{T}} = \frac{2P\,\cos\theta\, S(f)\,\sqrt{T}}{\Delta L(f)\, c} \tag{4.2}$$

The free parameters in Equation 4.2 are the laser power P and the integration time T. The angle of incidence θ is determined by the locations of the viewports on the vacuum tower. Hence, the angle of incidence is assumed to be $\theta = 35°$.

$$P(f) = \frac{c}{2\cos\theta\, S(f)} \frac{\Delta L(f)\, SNR}{\sqrt{T}} \tag{4.3}$$

Here, $P(f)$ represents the peak-to-valley power of the modulated signal. The minimum required laser power is given in Table 4.1 at various frequencies and integration times.

Table 4.1: Minimum required peak-to-valley laser power to achieve an SNR of 100

Frequency [Hz]	Integration time [s]	Laser power [mW]
10	1	0.07
100	10	0.87
500	10	15.1
1000	30	37.4

It is possible to use a less powerful laser when desired. However, the required integration time to reach the same SNR scales quadratically, so to cover the full bandwidth with reasonable integration times, the output power will need to be in the order of tens of mW.

4.4 Proposed photon calibrator concept

Laser source The source must be a 40 mW laser with a wavelength between 1350 and 1580 nm, excluding 1550 nm to avoid interference with the interferometer signal.

Transmitter / receiver The transmitter must house the laser source and optics to polarise the beam such that it is reflected by the test mass with s-polarisation. The beam must also be split into two parallel outgoing beams. A portion of the beam must be directed to a photodetector via a beam sampler to measure the incident power $\tilde{P}_i$. Power modulation can be achieved either by directly modulating the pump diode current or by using an external AOM. At the receiver module, both reflected beams must be redirected by a beam steering mirror onto a single photodetector to measure the total reflected power $\tilde{P}_r$. Further research is required to determine a definitive optical layout for the transmitter and receiver.

Beam path The proposed beam path is shown in Figure 4.8. The parallel Pcal beams enter and leave the cryogenic system through a series of windows within the shields. Further details of the design requirements for these windows are provided in chapter 5.

4.4.1 Calibration uncertainty budget

Several error sources and their expected uncertainty remain unchanged with respect to the derived values in chapter 3. The relationship between the modulated laser power $P(f)$ and the induced test mass displacement $z(f)$ is used as a reference to determine the error sources, which, in the case of total specular reflection, is given by

$$z(f) = -\frac{2\cos\theta}{c} \frac{1}{m(2\pi f)^2} P(f) \tag{4.4}$$

Because of the change of the source wavelength, the uncertainty on the reflectance ρ is not taken into account. Instead, this is captured by the uncertainty in the optical efficiency.

Considering the measured values of the power at the transmitter ($\tilde{P}_i$) and receiver ($\tilde{P}_r$), the power P is estimated as

$$P \approx \frac{\tilde{P}_i + \tilde{P}_r}{2} \tag{4.5}$$

The uncertainty associated with the optical efficiency $\eta = \tilde{P}_r/\tilde{P}_i$ is treated as a rectangular distribution. Although in practice this is determined by the measurements, it is now assumed that each window or viewport is AR-coated with a reflectance of $R < 0.1\,\%$ such that the efficiency $\eta = (1-R)^8$ in the case of eight windows. The estimated uncertainty on the efficiency is then given by $\sigma_\eta = (1-\eta)/2\sqrt{3} = 0.23\,\%$.

Table 4.2: Frequency-independent calibration uncertainty for the cryogenic windows Pcal

Parameter	**1σ uncertainty**
Angle of incidence	0.47 %
Mirror mass	0.20 %
Optical efficiency [a]	0.23 %
Test mass rotation	0.26 %
Photodiode calibration	0.17 %
Total	0.64 %

[a] In practice, in-situ measurements of the power determine the uncertainty

The most significant reduction in error is achieved by mitigating the effects of test mass deformation on the force-to-displacement response. The frequency-dependent uncertainty is evaluated for a split-beam configuration with an offset of $d = 45$ mm. This contribution is added in quadrature to the total uncertainty budget and plotted in Figure 4.11.

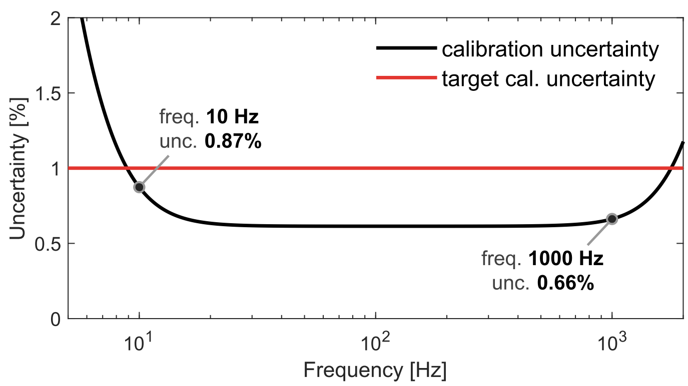


Figure 4.11: Frequency-dependent calibration uncertainty for the cryogenic windows Pcal

4.5 Conclusion

In this chapter, the requirements have been established for a photon calibrator to meet the desired SNR and calibration uncertainty. With the proposed photon calibrator system, the calibration uncertainty is expected to remain below 1 % between 10 Hz and 1 kHz. Moreover, the increased source power and altered wavelength ensure that an SNR of at least 100 can be achieved over this bandwidth. The concept requires the need for cryogenic windows.

5 Cryogenic windows

In chapter 3, it has been concluded that the performance of a photon calibrator using the optical lever is limited by the shield apertures that constrain the beam to be centred on the test mass. Several concepts were explored in chapter 4 to mitigate this limitation, and ultimately, the concept with cryogenic windows was proposed.

In this chapter, the feasibility of such cryogenic windows for ETpathfinder's thermal shields is investigated. An analytical model is constructed to analyse heat transfer between the windows and shields. Ultimately, a proposal for the window design is presented.

5.1 Thermal radiation shields

The test mass is surrounded by four double-walled thermal radiation shields, which assist in reducing its temperature to approximately 15 K. The three innermost shields are actively cooled, whereas the outer floating shield passively reduces the radiative heat load on the second shield. Figure 5.1 shows the expected equilibrium temperatures for each shield and the corresponding required cooling power.

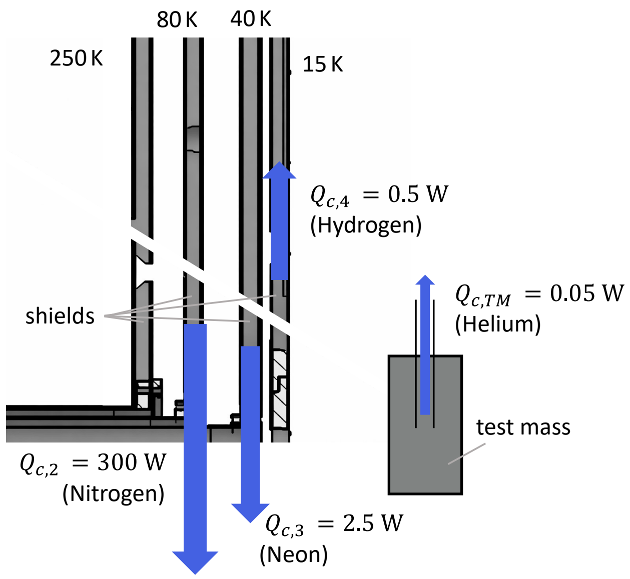


Figure 5.1: Equilibrium temperatures and total cooling capacity for each shield (the direction of the arrows is arbitrary)

The 80 K shield is cooled using a liquid nitrogen bath, which provides the largest cooling capacity. The 40 K and 15 K shields, as well as the payload, are cooled using a novel sorption-based cryocooler [28]. These sorption cells are designed to effectively cool the shields and the test mass while minimising vibrational noise from the compressors.

If windows were to be placed within the shields, the additional heat load from outside radiative heat to be absorbed by the window may not exceed 5 % of the cooling capacity of the shield in which this window is placed. This ensures that the current cryocoolers do not require a significant performance increase to accommodate these windows.

5.1.1 Window requirements

To transmit two Pcal beams at their optimal offset, each window must provide a clear aperture of 90 mm, motivating the use of windows with a total diameter of 100 mm. The window material must transmit the Pcal laser wavelength of 1460 nm, while reflecting or absorbing radiative heat from the environment and conducting any absorbed heat to the surrounding shields. Thermo-mechanical stresses arising from temperature gradients or from the mounting of the window must not lead to failure.

It is preferred that the windows also transmit visible light. This would enable viewing of the test mass with a camera, which would be advantageous for monitoring the interferometer beam and aligning the Pcal beams.

5.2 Radiation model

An analytical model is developed to compute the radiative heat transfer between the windows and shields, and is used to determine the incident radiation on the test mass through circular apertures in the shields. This model is also used to derive an explicit requirement on the maximum allowable heat load by computing the heat incident on the test mass for the current apertures. In this section, the key assumptions, parameters, calculations, and results of the model are presented.

5.2.1 Model assumptions

Several assumptions are made to keep the model concise while maintaining accuracy.

- The aluminium shields have an emissivity of $\varepsilon = 0.1$. Radiation that is not absorbed by the shields is diffusely reflected.
- All surfaces – shields, windows, test mass – behave as diffuse grey-body emitters.
- The shields are cooled to and kept at their equilibrium temperature.
- All shield apertures are assumed to be coaxial and equal in size.
- One side of the shield panels is modelled

5.2.2 Black-body radiation

All objects radiate heat Q to the environment according to the Stefan-Boltzmann law, given by the following expression.

$$Q = \varepsilon\sigma A T^4 \tag{5.1}$$

with the emissivity ε of the material, the area A of the emitting surface, the Stefan-Boltzmann constant $\sigma = 5.67 \cdot 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$, and the absolute temperature T in K.

The distribution of the emitted wavelengths λ by black-body radiation is given by the spectral exitance M_λ following Planck's law

$$M_\lambda = \frac{2\pi h c^2}{\lambda^5}\frac{1}{e^{\frac{hc}{\lambda k T}} - 1} \tag{5.2}$$

with the Planck constant $h = 6.63 \cdot 10^{-34}$ J, the speed of light $c = 299,792,458$ m/s, and the Boltzmann constant $k = 1.38 \cdot 10^{-23}$ J/K. Figure 5.2 shows the spectral exitance M_λ for the various temperatures of the emitting surfaces in units of W/m^2/μm.

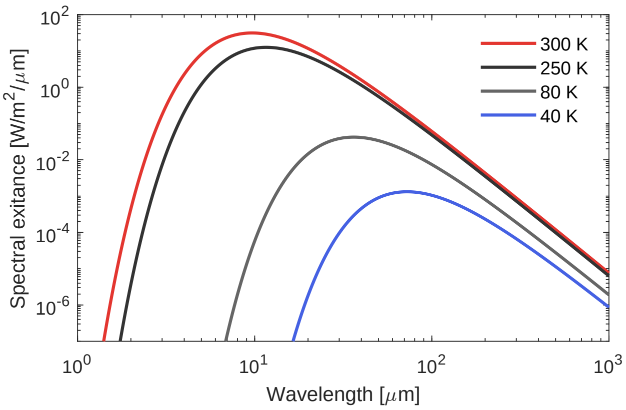


Figure 5.2: Spectral exitance of a black-body at various temperatures

Outer aperture as a diffuse emitter

A small aperture in a large enclosure can be treated as a diffuse emitter at the enclosure temperature, because the radiation exiting the aperture has been reflected multiple times, randomising its direction. The same holds for the aperture in the outer radiation shield, which receives randomly oriented radiation emitted from and reflected by the vacuum tower. Hence, the outer aperture is modelled as a black body at $T = 300$ K, emitting heat in the direction of the test mass at

$$Q_a = A_a \sigma T^4 = 3.61 \quad \text{W} \tag{5.3}$$

Diffuse emitters significantly simplify the computation of radiative heat transfer, as they allow heat exchange to be expressed analytically in terms of view factors.

5.2.3 View factors

The radiative heat transfer between two diffuse surfaces A_i and A_j depends on their emissivities, their relative temperatures, and the view factor F_{ij}, denoting the fraction of the emitted radiation leaving A_i that reaches A_j.

Calculating the view factor between two arbitrary geometries can, in most cases, only be achieved numerically. Analytical expressions have been derived for only a small number of simple geometries. For instance, the view factor between two coaxial circular surfaces with radius R at distance H, given by

$$F_{cc} = 1 + \frac{H^2}{2R^2} - \frac{H\sqrt{H^2 + 4R^2}}{2R^2} \tag{5.4}$$

This view factor is the same for both surfaces, since they are of equal size and symmetrically positioned. For modelling the radiation, $F_{\circ\circ}$ can be used directly to describe the view factor between two windows. Next, two important relations are introduced to determine the remaining view factors needed for the model.

Summation and reciprocity

In an enclosure, all radiation leaving a surface must reach any other surface in the enclosure. No radiative heat leaves the system, so that the sum of all the view factors related to the radiation leaving surface i, including F_{ii}, must equal 1. This is called the summation rule.

$$\sum_{j=1}^{n} F_{ij} = 1 \tag{5.5}$$

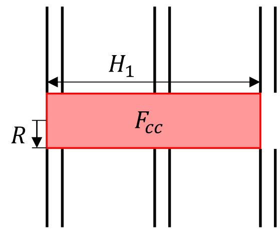
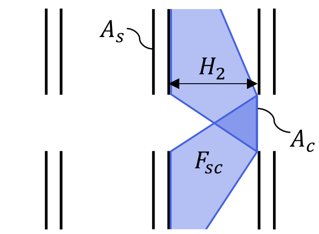
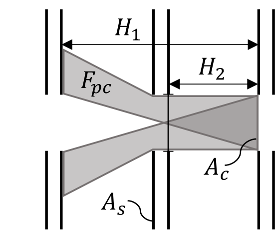
Another essential relation between the view factor and the emitting surface area is called reciprocity. This relation arises from the symmetry of radiative heat transfer between two surfaces, and is given by:

$$A_i F_{ij} = A_j F_{ji} \tag{5.6}$$

View factors for the model

The view factors between the shields and the apertures can be determined analytically using F_{cc} and the aforementioned relations. In Table 5.1, the expressions, as well as visualisations of the parameters, are shown.

Table 5.1: The various view factors used in the radiation model

Description	Schematic	Expression
View factor between two circular apertures with radius R at a distance H_1		$F_{cc} = 1 + \frac{H_1^2}{2R^2} - \frac{H_1\sqrt{H_1^2+4R^2}}{2R^2}$
View factor from a shield with area A_s to a circular aperture with area A_c at a distance H_2		$F_{sc} = \frac{A_c}{A_s}\left[1 - F_{cc}(H_2, R)\right]$
View factor from a shield with area A_s through an aperture onto the next aperture, with A_c, at distance H_1 The distance between apertures is H_2		$F_{pc} = \frac{A_c}{A_s}\left[F_{cc}(H_2, R) - F_{cc}(H_1, R)\right]$

It is assumed that all radiation emitted by a shield is incident on either the other shield or the window or aperture within that shield, i.e., the view factor from one face of the shield to another face of the same shield, at the corners, is negligible. It is verified using a Matlab view factor calculation tool [29] that these view factors are less than 0.02.

5.2.4 Radiative heat transfer

Now the means have been established, the radiative heat exchange between two surfaces i and j can be calculated using the following formula.

$$Q_{ij} = \frac{e_{b,i} - e_{b,j}}{\sum resistances} \tag{5.7}$$

where e_b is the blackbody radiation per unit area in W m^{-2} of the surface. The resistances can be divided into two categories: the surface and space resistance.

$$R_{surface,i} = \frac{1-\varepsilon_i}{A_i \varepsilon_i} \quad \text{or} \quad R_{space,ij} = \frac{1}{A_i F_{ij}} \tag{5.8}$$

The *surface* resistance represents how much a surface resists emitting radiation compared to an ideal blackbody, governed by the emissivity ε. The *space* resistance represents how difficult it is for radiation to travel between surfaces due to their relative position and orientation, captured by the view factor F_{ij} and the area A_i. Altogether, the expression for the radiation between two surfaces is given by

$$Q_{ij} = \frac{\sigma\left(T_i^4 - T_j^4\right)}{\frac{1-\varepsilon_i}{A_i\varepsilon_i} + \frac{1}{A_i F_{ij}} + \frac{1-\varepsilon_j}{A_j \varepsilon_j}} \tag{5.9}$$

5.2.5 Radiation model outline

The geometric parameters of the model are gathered from ETpathfinder's CAD drawings, such as the area of the front panels of the four shields given by A_i for $i \in \{1,2,3,4\}$, and the distances between shields and the test mass. The shields are assumed to be at their equilibrium temperature T_i, while the temperature of the vacuum tower is set to $T_0 = 300$ K, slightly above room temperature to provide a margin.

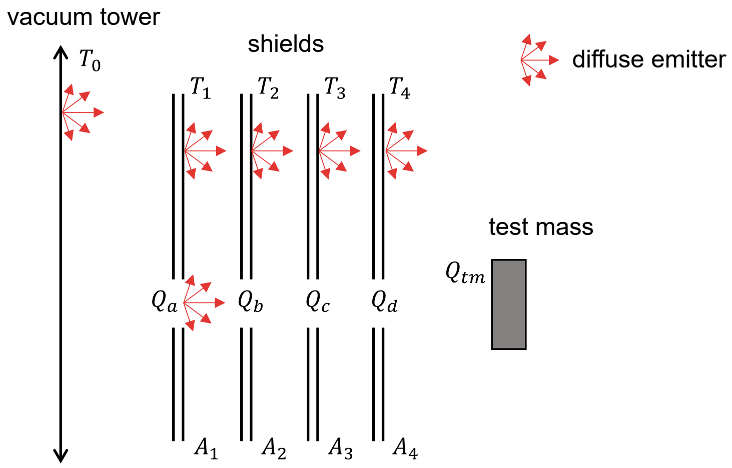


Figure 5.3: Visualisation of the analytical model parameters

The values to be computed are Q_b, Q_c and Q_d, as shown in Figure 5.3. These represent the net radiative heat flow through the aperture, in W. Additionally, the net amount of radiation incident on the test mass is given by Q_{tm}. Comparisons with numerical results indicate that deriving an analytical expression for the view factors from the apertures onto the test mass is non-trivial. Hence, the value of Q_d with $Q_d \geq Q_{tm}$ is taken as the heat load of interest.

Maximum allowable heat load

Apertures with a diameter of 20 mm are currently present in the shields to provide access to the test mass for the optical levers, resulting in a heat load of

$$Q_{d,20} = 1.27 \quad \text{mW} \tag{5.10}$$

The goal is to achieve a similar resulting heat load with the addition of the cryogenic windows. Hence, a requirement on the maximum allowable heat load is set to **1.3 mW**.

5.2.6 Validation with numerical simulations

Numerical simulations are performed to verify the results of the radiative heat transfer model. A 3D model of the shields is created with a series of coaxial apertures in the front panels (on the left in Figure 5.4). The shields are assumed to be cooled to $T_i = \{250, 80, 40, 15\}$ K, and the environment is at $T_0 = 300$ K. The radiative heat transfer module in COMSOL 5.6 is used to create and simulate the model. Additional details and results of this validation study are provided in the Appendix, section C.1.

View factor validation

COMSOL allows the computation of view factors for a given geometry via numerical integration over the mesh elements. Hence, the analytically derived view factors are compared with the corresponding numerical values in Table 5.2.

Table 5.2: Analytical and numerical values of the view factors (D = 100 mm)

View factor	Matlab value [-]	COMSOL value [-]
F_{aa}	1	0.99
F_{ab}	0.38	0.38
F_{ac}	0.17	0.17
F_{ad}	0.10	0.10

From these results, it can be concluded that the analytical expressions for these view factors are correct. The numerical COMSOL values converge to the analytical Matlab values as the mesh is increasingly refined.

Radiation validation

The ambient and mutual radiation at the apertures Q_a, Q_b, Q_c, and Q_d are presented in Table 5.3. The *ambient* radiation is the radiation from the room temperature vacuum tower entering through the apertures, whereas the *mutual* radiation is the radiation exchanged with all other surfaces, which in this case are the shields.

Table 5.3: Analytical and numerical values of ambient and mutual radiation (D = 100 mm)

	Ambient radiation [W]		Mutual radiation [W]	
Radiation	**Matlab value**	**COMSOL value**	**Matlab value**	**COMSOL value**
Q_a	3.61	3.57	0	0
Q_b	1.34	1.36	1.01	0.96
Q_c	0.62	0.61	0.38	0.45
Q_d	0.34	0.36	0.18	0.21

The ambient radiation values also converge to the analytically determined values as the mesh is refined. This is expected since these values are governed by the view factors, although they are computed differently.

The analytical values of the mutual radiation agree with the numerical values to within 10-30%. This discrepancy arises because the analytical model considers only mutual radiation between the previous shield and the shield before that, whereas in fact, more surfaces contribute. Moreover, the shield reflectivities are not accounted for in the Matlab model. Nonetheless, the total discrepancy between the two values is considered acceptable.

Staggered and coaxial

Both models in the previous section assume a series of coaxial circular apertures, whereas in fact the apertures are staggered by 35° due to the beam's incidence angle. This assumption is made because no elementary analytical expression exists for the view factor between discs with a lateral offset. In COMSOL, a model of the radiation shields is created with staggered apertures (on the right in Figure 5.4), of which the resulting radiation values are compared to the coaxial model are presented in Table 5.4.

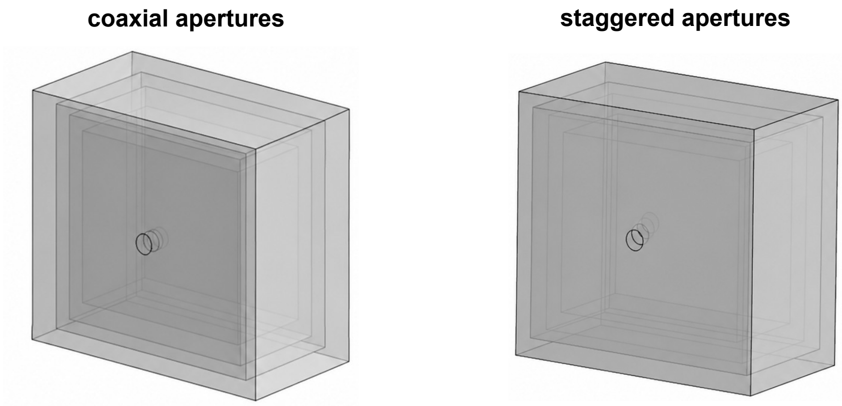


Figure 5.4: Models used to compare the effects of coaxial and staggered apertures

Table 5.4: Radiation values for coaxial and staggered apertures (D = 100 mm)

Total radiation	Coaxial [W]	Staggered [W]
Q_a	3.57	3.57
Q_b	2.32	2.26
Q_c	1.10	0.99
Q_d	0.66	0.56

The total radiation, assuming coaxial apertures, always exceeds that of the staggered configuration by 0-20%. Therefore, this assumption is incorporated into the analytical model, since it provides a conservative estimate of the heat load.

5.3 Window material selection

Cryogenic windows must transmit the 1460 nm laser light while effectively reflecting or absorbing thermal radiation. The absorbed heat must not raise the window temperature by too much to restrict the amount of heat radiated toward the test mass. Moreover, thermally induced stresses within the window must not lead to failure. In this section, several candidate window materials are selected in accordance with these principles.

5.3.1 Transmissivity

Based on their transmission ranges, sapphire, fused silica, N-BK7, and crystalline quartz are selected as suitable window materials. This is because their cut-off wavelengths lie in the mid- to far-IR region (see appendix Figure C.1). The transmissivity τ of a material is the fraction of the incident light that is transmitted. The exact value of τ is dependent on the material's spectral transmissivity τ_λ and the spectral irradiance E_λ given by Planck's law (5.2).

$$\tau(T) = \frac{\int_0^\infty \tau_\lambda(\lambda) E_\lambda(\lambda, T)\, d\lambda}{\sigma T^4} \tag{5.11}$$

Here, T denotes the temperature of the black-body source that irradiates the surface.

Amorphous materials

Fused silica and N-BK7 have a single transmission band with a relatively low cut-off wavelength around 3 μm, and their transmission spectra are independent of the material temperature. These materials have an amorphous (non-crystalline) structure.

Due to the low transmission cut-off, fused silica absorbs most of the 300 K radiation as a outer window and effectively all of the 80 K radiation as an inner window. Table 5.5 shows the values of the total transmissivity τ of various incident spectra, calculated using (5.11).

Table 5.5: Transmissivity of fused silica for different radiation temperatures

$\mathbf{T_{rad}}$	**300 K**	**250 K**	**80 K**	**40 K**
τ	0.001	0.0002	0	0

The transmission of radiation from cryogenic surfaces ($\leq$ 80 K) is negligible. Overall, fused silica and N-BK7 show excellent transmission properties as inner or outer windows.

Crystalline materials

Crystalline materials such as sapphire and crystalline quartz have a second transmission band in the extreme-IR region (see appendix Figure C.1). Their spectral transmissivity depends on the material temperature, such that at lower window temperatures, the second transmission band broadens.

Table 5.6 shows the transmissivity τ of sapphire at various temperatures of the window T_{window} and for various incident spectra, indicated by the black-body temperature T_{rad}.

Table 5.6: Transmissivity of sapphire at different window and radiation temperatures

$\mathbf{T_{window} \backslash T_{rad}}$	**300 K**	**250 K**	**80 K**	**40 K**
250 K	0.04	0.02	0.05	0.18
80 K	0.05	0.04	0.20	0.38
40 K	0.07	0.06	0.33	0.48
15 K	0.09	0.09	0.42	0.54

If sapphire windows were to be placed in the inner three shields, approximately 8 mW of heat emitted by the shields or by previous windows would be transmitted to the test mass. They may be used as an outer window, but only in combination with an inner window made of a different material. Otherwise, the transmitted ambient radiation will result in 12 mW of incident heat on the test mass.

The transmitted spectra for various materials, along with the transmissivity values for N-BK7 and crystal quartz, are shown in the appendix section C.1.

5.3.2 Thermo-mechanical properties

Thermal and mechanical material properties are used in the model to determine thermal gradients and stresses within the windows. In Table 5.7, a selection of these properties is presented for the four candidate window materials.

The thermal conductivity k and expansion coefficient α are strongly dependent on temperature. In the conduction model presented in section 5.4, this is accounted for when computing the window temperature and thermal stresses. Some properties of crystalline materials depend on the orientation of the crystal axes; however, the model assumes an isotropic material, using the least ideal value for each property.

Table 5.7: Optical and thermo-mechanical properties of different window materials

Property	**N-BK7**	**Fused silica**	**Sapphire**	**Crystal quartz**
Transmission band [μm]	0.3 – 2.8	0.2 – 3.0	0.2 – 5.5 100 – 500	0.2 – 3.5 40 – 500
Suitable as inner window?	Yes	Yes	No	No
Suitable as outer window?	Yes	Yes	Yes [a]	Yes [a]
Thermal conductivity at 80 K [W/m K]	0.6	0.9	800	100
Thermal expansion coefficient at 80 K [ppm/K]	0.68	- 0.70	0.51 (c-axis)	0.92 (c-axis)
Young's modulus [GPa]	82	73	435 (c-axis)	97 (c-axis)

[a] when used in combination with a different material inner window

At cryogenic temperatures, the thermal conductivity of the crystalline materials is much higher than that of the amorphous materials. Consequently, the amorphous materials are expected to exhibit larger temperature gradients. On the other hand, the thermal expansion coefficients of these materials are relatively similar and remain below 1 ppm/K. As a result, the thermal stresses may remain small, even in the presence of large temperature gradients.

To obtain a clear understanding of the thermal gradients and stresses in these materials, an analytical model of the window conduction is developed.

5.4 Window conduction

In section 5.2, an analytical model has been created to determine the radiative heat load on a window placed within the thermal shields. In section 5.3, a selection of candidate window materials is made on the basis of their transmissivity and thermo-mechanical properties.

In this section, the conductive heat transfer within the window and to the shields will be discussed using an analytical model that uses the radiation model output to compute the conductive heat transfer through the window for the various materials considered.

5.4.1 Heat equation

To model the window conduction analytically, the general heat conduction equation in cylindrical coordinates is used.

$$\frac{1}{r}\frac{\partial}{\partial r}\left(k\,r\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p\frac{\partial T}{\partial t} \tag{5.12}$$

The right-hand side of the equation vanishes because the steady-state solution is of interest. In addition, the angular term is zero because of axisymmetry, and the axial term is zero under the thin window assumption. Figure 5.5 a schematic of the radiative heat load, heat sinks, and boundary conditions.

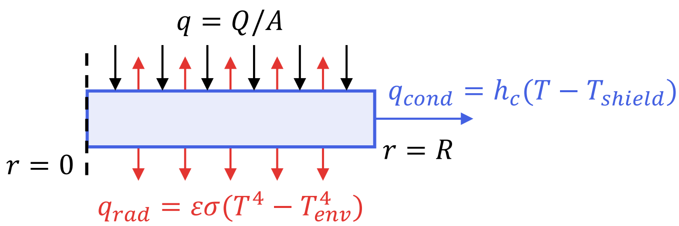


Figure 5.5: Schematic of the axisymmetric window conduction problem

The volumetric heat flow $\dot{q}$ is written in terms of incident radiation Q, and the window volume $V = \pi R^2 t$. Moreover, the linearised term $h_{rad} = 4\varepsilon\sigma T_{env}^3$ is included to account for the outward radiation emitted by the window as it heats up, where T_{env} denotes the environmental temperature. Altogether, the heat equation is given by

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) - \lambda^2(T - T_{env}) = -\frac{Q}{\pi R^2 k t} \tag{5.13}$$

with

$$\lambda^2 = \frac{2h_{\text{rad}}}{kt}. \tag{5.14}$$

In the Appendix section C.4, the full derivation of the radial temperature distribution $T(r)$ is presented, resulting in the following equation:

$$T(r) = T_0 + \frac{Q}{\pi R^2 h_{rad}}\left[1 - \frac{h_c I_0(\lambda r)}{k\lambda I_1(\lambda R) + h_c I_0(\lambda R)}\right] \tag{5.15}$$

with and I_1 and I_0 the first- and zero-order modified Bessel function of the first kind.

This expression represents the steady-state radial temperature profile in a thin window of radius R and thickness t, subject to a uniform total heat load Q, with radiative cooling on both faces (linearized with h_{rad}), and edge conduction through contact conductance h_c.

Validation with numerical simulations

The analytic solution is validated against a numerical model of the window under a load Q, about which more details are presented in the Appendix section C.3. It is concluded that the solution for $T(r)$ is accurate within 5 % for temperature gradients of less than 20 K. This error arises from the linearised term h_{rad}, and increases for larger gradients.

5.4.2 Thermal stress

Assuming the windows are mounted stress-free, temperature gradients will induce the dominant stresses within the material. For a radial temperature distribution $T(r)$ in a disc, the radial σ_r and tangential σ_θ stresses are given by

$$\sigma_r = (\alpha E)\left[\frac{I}{R^2} - \frac{1}{r^2}\int_0^r T(r)\, r\, dr\right] \tag{5.16}$$

$$\sigma_\theta = (\alpha E)\left[\frac{I}{R^2} + \frac{1}{r^2}\int_0^r T(r)\, r\, dr - T(r)\right] \tag{5.17}$$

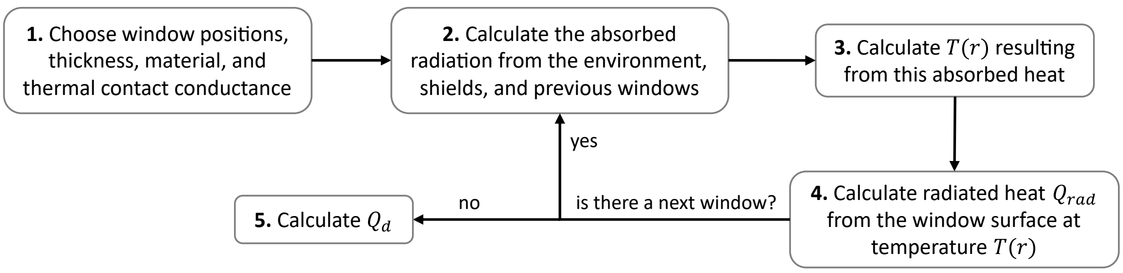
with α the coefficient of thermal expansion, and E the Young's modulus of the material. I is a constant given by

$$I = \int_0^R T(r)\, r\, dr \tag{5.18}$$

These equations are adapted from [30] and verified with numerical results. In the following section, these equations will be used to determine the expected thermal stresses within the cryogenic windows for various configurations.

5.5 Cryogenic window model results

Using the equation for the radial temperature profile (5.15) and the Q_i values obtained from the radiation model (section 5.2), a combined analytical model is constructed to compute the temperature distributions and thermal stresses of the cryogenic windows. The complete model is provided in the Appendix. In short, the model consists of the following steps:



The radiated heat from the window surface Q_{rad} expressed in units of W is given by the following expression and is computed numerically in Matlab.

$$Q_{rad} = 2\pi\varepsilon\sigma\int_0^R \left(T(r)^4 - T_n^4\right) r\, dr \tag{5.19}$$

With T_n the initial temperature of the next window. Steps 2 through 4 are computed sequentially for each window, and the radiation Q_d is computed at the end. The model is used to analyse various combinations of window materials, window positions, and the number of windows, of which a selection is presented in this section.

5.5.1 Window locations

If the heat load requirement can be met with only a small number of windows, this would be a preferred solution over placing a window in every shield. Therefore, it is investigated which shields definitely require a window based on first-principle calculations.

Outer window The outermost window must absorb the majority of the radiative heat from the environment. In section 5.1, it was stated that the heat absorbed by a window may not exceed 5 % of the total cooling capacity of the shield it is placed in. This excludes the 40 K and 15 K shields as inner window locations, as these windows would be subjected to heat loads of 0.52 W and 0.34 W, respectively. The 80 K shield can be used for the outer window, since the incident heat of 1.3 W amounts to less than 1 % of the shield's total cooling power. The 250 K shield is not actively cooled, but it can hold an outer window, only in combination with a subsequent 80 K window.

Inner window The temperature of the innermost window determines the radiated heat towards the test mass. It has been concluded in subsection 5.2.5 that the radiation Q_d must be less than 1.3 mW. This excludes the 250 K and 80 K shields as outer-window locations, because, in the ideal case in which a window is uniformly cooled to the shield temperature, the radiated heat would already exceed the requirement. Thus, the inner window must be placed within the 40 K or 15 K shield, resulting in radiation Q_d of 0.89 mW and 0.02 mW respectively, assuming an ideal window.

5.5.2 Two-window configurations

From the previous section, it follows that at least two shields must each hold a window. If just these two windows are considered, the outer window must be placed in the 80 K shield, while the inner window can be placed in the 40 K or 15 K shield. In this study, both configurations are investigated using the model.

It has been determined that fused silica is the best material for both windows. Fused silica has a lower thermal conductivity than sapphire or crystal quartz but a more favourable transmissivity, which provides greater heat absorption. The transmitted radiation by crystalline materials exceeds the re-radiated heat from the amorphous materials. NBK-7 performs worse than fused silica, because its thermal conductivity is lower.

80 K – 15 K

With fused silica as the outer window in the 80 K shield, the window heats up. Its temperature distribution depends on the thermal conductivity k, the window thickness t, and the thermal contact conductance h_c between the window and the shield. This heating of the outer window can result in an excessive heat load on the inner window, which, in this case, is positioned in the 15 K shield.

In Figure 5.6, the incident radiation onto the inner window is plotted as a function of h_c and t of the outer window. The red line indicates 5 % of the total cooling capacity of the inner shield, representing the maximum allowable heat load of this window.

It is difficult to specify an expected value for the contact conductance h_c, as it strongly depends on factors such as surface roughness, materials, temperature, and contact pressure. Siddappa showed in [31] that in similar vacuum cryogenic conditions, the value for h_c can

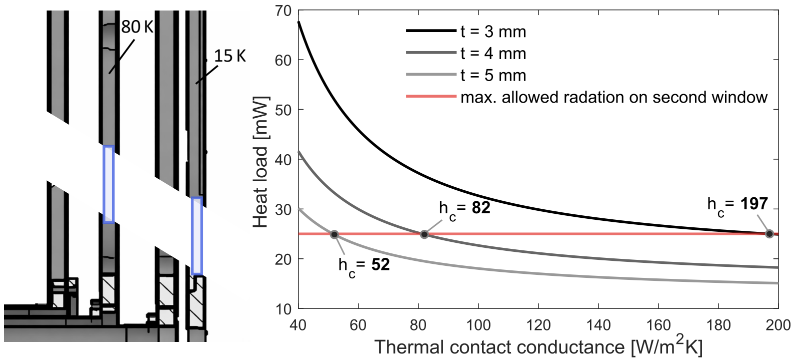


Figure 5.6: Incident heat from the 80 K window onto the inner 15 K window as a function of the contact conductance for various thicknesses t of the outer 80 K window

range from 10 to over 1000 W/m^{2}K. Hence, it is necessary to determine the contact conductance experimentally under representative conditions, when possible.

80 K – 40 K

Another possible combination of two windows is shown on the left in Figure 5.7. In this case, the maximum allowable heat load on the second window is not exceeded. In the middle of Figure 5.7, the resulting window temperature profiles are plotted, as well as the principal thermal stresses on the left calculated with (5.16) and (5.17).

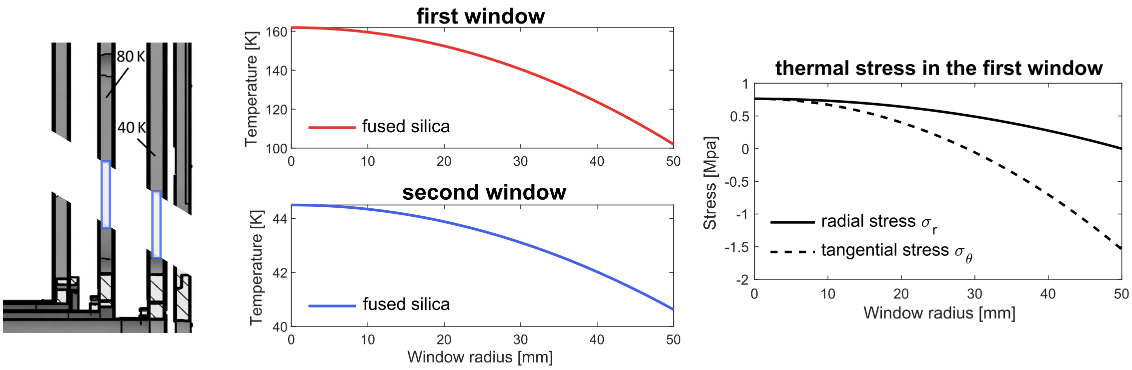


Figure 5.7: Temperature profiles and thermal stress of a a 80 K outer window and 40 K inner window for $h_c = 100$ W/m^{2}K and $t = 3$ mm

It is generally difficult to tell whether glass will fail under a certain stress because failure often originates from microcracks within the material. However, with a peak-to-peak stress of 2.3 MPa, this window is unlikely to fail. That is, under the assumption that the window is mounted on a compliant structure so that the thermal expansion mismatch with the shield does not induce a significant stress. To determine whether this configuration can achieve the desired heat load Q_d, the model is used to compute it. In Figure 5.8 this is plotted for various window thicknesses t and h_c for both windows.

Among the two-window combinations, the 80 K – 15 K configuration with at least 5 mm-thick windows fused silica windows is preferred, as it requires the lowest thermal contact conductance, approximately 52 W/m^{2}K. However, it is unclear if this thermal contact is achievable in ETpathfinder's cryogenic and vacuum environment, so a three-window configuration with a less strict requirement on h_c is preferred. This is investigated further in subsection 5.5.3.

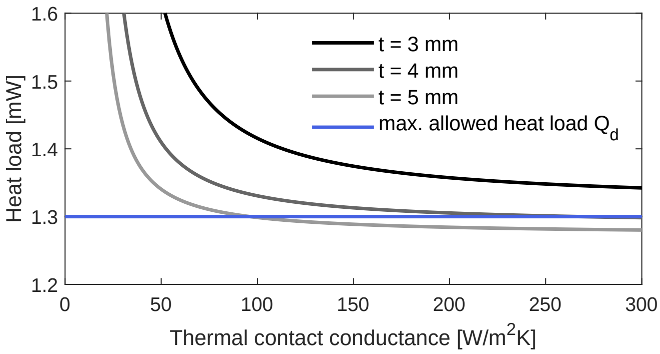


Figure 5.8: Heat load Q_d as a function of the contact conductance for various thicknesses of a 80 K outer window and a 40 K inner window

Stress birefringence

If a transparent, isotropic material such as fused silica is subjected to stress, it can become anisotropic, resulting in a direction-dependent refractive index. This is called stress birefringence, and it can cause a phase shift of the transmitted light, which is dependent on the stress-optic coefficient $C = 3.5 \cdot 10^{-12}$ Pa^{-1} for fused silica, window thickness $t = 5$ mm, and the Pcal wavelength of $\lambda = 1460$ nm. Resulting in a phase retardation δ of

$$\delta = \frac{2\pi}{\lambda} C\, t\, (\sigma_r - \sigma_\theta) = 0.094 \quad \text{[rad]} \tag{5.20}$$

If the linear polarisation axis were oriented at 45° with respect to the principal stress axes, this would result in a slightly elliptical polarisation with a ratio of the p- to s-polarisation components of 0.047. The resulting variation in the reflectivity ρ of the Pcal beam on the test mass is $\sigma_\rho \approx 0.03$ % considering the worst case. Overall, the effect of stress birefringence on the Pcal calibration uncertainty is negligible.

5.5.3 Three-window configurations

It has been shown that with two windows, the heat load can be reduced to less than 1.3 mW. Under the condition that a thermal contact conductance of $h_c \approx 100$ W/m^{2}K is achievable. Given the cryogenic and vacuum conditions of this system, and the fact that the window must be mounted without excessive stress, the value of h_c may likely be in the tens of W/m^{2}K.

80 K – 40 K – 15 K

The resulting heat load is computed for various configurations and window materials using three windows. Of those considered, three fused silica windows in the 80, 40, and 15 K shields have been shown to be best at maintaining a low radiative heat load. Figure 5.9 shows the heat load Q_d as a function of contact conductance and window thickness, which are considered to be the same for each window.

Above $h_c = 20$ W/m^{2}K, the heat load is minimum for the three considered cases, at which point the remaining 3.8 mW is the transmitted radiation. At a lower h_c, the heat load increases steeply because the windows heat up and subsequently radiate more heat themselves. Part of the reason this increases so abruptly is that all windows in the model are assumed to have the same contact conductance, which adds up their effects.

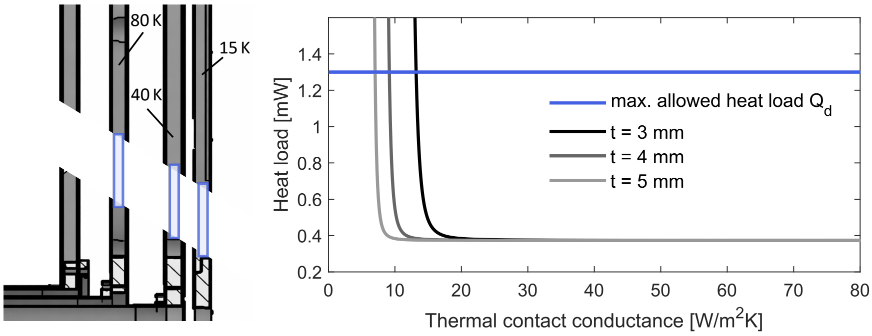


Figure 5.9: Heat load Q_d as a function of the contact conductance for various thicknesses of an 80 K outer window and a 40 K inner window

5.5.4 Proposed window configuration

From the considered window configurations, the 80 K – 40 K – 15 K combination with fused silica windows is proposed. It has shown the greatest reduction in heat load among the three window configurations, while requiring a relatively low contact conductance of $h_c > 20$ W/m^{2}K. In theory, the desired heat load can be achieved with two windows, but this depends strongly on whether a high thermal contact conductance of $h_c > 100$ W/m^{2}K can be achieved. Therefore, using three windows is preferred for added security.

5.5.5 Mounting the windows

A critical aspect of implementing these cryogenic windows is how they are mounted to the shield. It is important to provide a good thermal connection to ensure the windows are properly cooled while preventing excessive stresses in the glass caused by the relative thermal contraction of the shield at low temperatures.

In the Appendix section C.6, a qualitative study is presented on the effects of various cooling and clamping methods on the heat distribution in the window. Although the results are not directly related to or comparable to the rest of this thesis, they may provide insights for further research into effective cooling or mounting of the windows.

5.6 Conclusion

In this chapter, the feasibility of cryogenic windows as part of a Pcal system for ETpathfinder has been analysed. To determine the radiative heat loads on and between windows, an analytical model has been constructed. The maximum allowable heat load through the windows is 1.3 mW, as this equals the heat load through the currently present apertures. The temperature distribution of the windows under a radiative load has been computed using analytical and numerical models. The thermal stress arising from the temperature gradient in the window is not expected to cause failure, provided that the window mount does not constrain the thermal expansion mismatch between the window and the shield or mount.

A configuration of three fused silica windows in the 80, 40, and 15 K shields is proposed and expected to reduce the radiative load to less than 0.5 mW, assuming that the thermal contact conductance between the windows and shields exceeds 20 W/m^{2}K. Obtaining an estimate for h_c requires further research and analysis on the design of a cryogenic window mount.

6 Conclusion and recommendations

In this thesis, the analysis and conceptual design of a photon calibration system for ETpathfinder, the cryogenic gravitational wave detector testbed, were presented. An initial research goal was set to **conceptualise a photon calibration system for ETpathfinder with a calibration uncertainty under 1 % across a 10 Hz to 1 kHz bandwidth**.

Ultimately, two conceptual systems have been proposed: a single-beam photon calibrator with limited performance, and a two-beam photon calibrator with cryogenic windows that meets the requirement across the full bandwidth.

6.1 Conclusion

The first concept is intended as a simple, cost-effective system that utilises part of the already-present optical lever system. In this system, the optical lever also serves as a Pcal beam reflected from the test mass centre. Numerical simulations have shown that this single-beam configuration induces local surface deformation, measured by the interferometer as an effective displacement. Consequently, the calibration accuracy is limited at higher frequencies. Power-modulation of the 674 nm SLD source is not possible using its current module, since the higher-order harmonics of the modulation signal are not sufficiently suppressed. This system is expected to have a sub-percent calibration uncertainty up to 180 Hz.

To meet the target calibration uncertainty requirement, a second conceptual system has been proposed. This design uses a 1460 nm laser source to maximise reflection from the test mass coating. This source has an output power of 40 mW, to achieve an SNR $\geq$ 100 over the full bandwidth. Two parallel Pcal beams positioned 45 mm from the test centre are used to reduce the uncertainty due to local deformation to less than 0.2% at 1 kHz. Cryogenic windows must be placed within the shields to provide optical access for the beams.

The windows absorb radiative heat from the environment and conduct it away toward the shields, while transmitting the Pcal beams at 1460 nm. It is determined that the maximum allowable heat load on the test mass is 1.3 mW. Fused silica as the window material is found to best reduce the heat load due to its favourable transmission range. Its relatively low thermal conductivity leads to a temperature gradient in the window, but due to the low thermal expansion at cryogenic temperatures, the resulting thermal stress is approximately 1.5 MPa and is not expected to cause failure nor affect the Pcal performance. A proposal is made to configure three windows in the 80 K, 40 K, and 15 K shields to reduce the heat load on the test mass to less than 0.5 mW.

With the cryogenic windows included in the two-beam conceptual photon calibration system, the total calibration uncertainty is expected to be less than 0.87% over the full modulation range from 10 Hz to 1 kHz.

6.2 Recommendations

The work in this thesis has provided a foundation for implementing a photon calibration system at ETpathfinder. Several recommendations are made for future work on this topic.

For the optical lever photon calibrator, a suitable method for modulating the laser power must be selected. If the reflectance of the SLD source on the test mass coating cannot be determined experimentally with sub-per cent uncertainty, a new source should be used at a highly reflective wavelength (1350-1580 nm) and with greater output power (>40 mW). It must also be tested experimentally how the modulated power affects the optical lever positional measurements. For both proposed photon calibrator concepts, a more detailed system must be designed, including the optical layouts and components of the transmitter and receiver modules.

The cryogenic windows require further research before implementation is realised. The theoretical results in this thesis must be verified experimentally, including the radiative heat loads from apertures and conductive heat transfer in a window. Additional methods can be studied to reduce the radiative heat load, e.g. by using baffles between apertures to block and absorb mutual radiation from the shields. The effects of locally varying the shield emissivity must be studied, for example, using black coatings. Lastly, the design and analysis of a window mount is required to effectively cool the window while avoiding excessive thermal stress within the window.

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A COMSOL simulations and parameters

In this appendix, a brief description of the methods and parameters of the COMSOL simulation is provided. To access all the code and files used for and referenced in this thesis, a repository is created at `https://github.com/larskooy01/master-thesis`.

All simulations consider the following parameters unless stated otherwise.

Table A.1: COMSOL model parameters

Parameter	Variable	Value
Mirror radius [mm]	`r`	75
Mirror thickness [mm]	`t`	80
ITF beam radius [mm]	`w0`	2.2
Pcal beam radius [mm]	`w1`	0.5
Pcal force [N]	`F`	1
Young's modulus [Gpa]	`E`	155
Poisson's ratio [-]	`nu`	0.28
Density [kg/m^3]	`rho`	2329

A.1 Eigenmode analysis of test mass

This analysis is conducted using a 3D model of the test mass and using the Eigenfrequency Study in COMSOL. No forces or constraints are set to any of the faces. With the following settings.

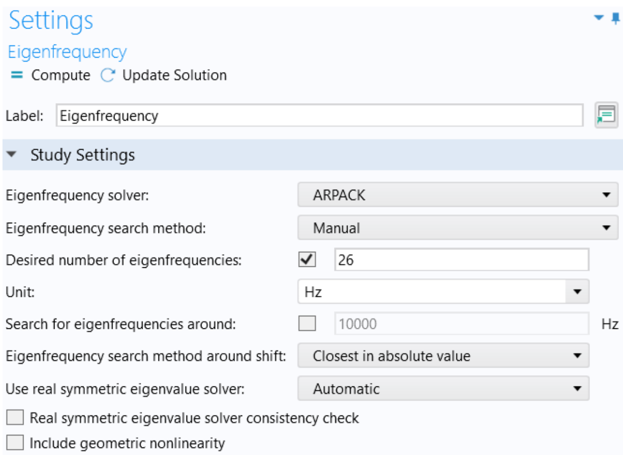

Figure A.1: Settings of the eigenfrequency study

A.2 Local deformation – single beam

This study uses a 2D axisymmetric model with a fine mesh of the top surface and the centreline. The Pcal force is taken as 1 N, which makes it convenient for calculating the effective deformation in m/N.

Within the structural mechanics module, a boundary load is modelled on the test mass front surface in the z-direction, expressed as:

```
-p*exp(-2*r²/w1²)
```

The back surface of the mirror is fixed to represent the inertia. First, the results in [16] are reproduced with the parameters from that paper to verify the model settings. A Stationary Study is used.

The resulting z-displacement field of the front surface taken over the radius, indicated by the variable w, is exported to Matlab, where the effective displacement is calculated:

```
% Main beam profile
w0 = 2.2e-3;                % waist radius
I = exp(-2*(r.^2)/w0^2);    % intensity profile

% Compute normalisation factor kI
num_I = 2*pi*trapz(r, r .* I);
kI = 1 / num_I;

% Compute D_eff
D_eff = 2*pi * kI * trapz(r, r .* I .* w)
```

A.3 Local deformation – two beams

For this study, the 3D model is used. An additional variable d is introduced to represent the Pcal beam offset between the two beams. On the test mass front surface, two spots are modelled as a single boundary load of each 0.5 N, in the z-direction expressed as:

```
-p* ( exp(-2*(x²/w1²+(y-d)²/w1²)) + exp(-2*(x²/w1²+(y+d)²/w1²)) )
```

Around the spot locations, a finer mesh is used to accurately resolve the deformation. A Stationary Study is used.

The resulting z-displacement field of the front surface area, indicated by the variable w, is exported to Matlab, where the effective displacement is calculated:

```
% Distribute data on uniform grid
xlin = linspace(min(x), max(x), nx);
ylin = linspace(min(y), max(y), ny);
[X, Y] = meshgrid(xlin, ylin);
W = griddata(x, y, w, X, Y, 'natural');
R = sqrt(X.^2 + Y.^2);

% Main beam profile
w0 = 2.2e-3;                 % waist radius
```

```
I = exp(-2 * (R.^2) / w0^2); % intensity profile

% Determine area element
dx = X(1,2) - X(1,1);
dy = Y(2,1) - Y(1,1);
dA = dx * dy;

% Compute normalisation factor kI
kI = 1 / (sum(I(:)) * dA);

% Compute D_eff
D_eff = kI * sum( I(:) .* W(:) * dA, 'omitnan' )
```

A.4 Bulk deformation – two beams

The same model, mesh and parameters are used as presented in section A.3. However, the boundary loads are now modelled as a harmonic perturbation at a constant frequency f, and the back surface of the test mass is not constrained.

In this analysis, the Frequency Domain Study is used to simulate a modulated load at a range of frequencies f. An auxiliary sweep is taken of the offset parameter d to obtain results for various offsets in the vicinity of the nodal circle of the drumhead mode. The study settings are displayed in the figure below.

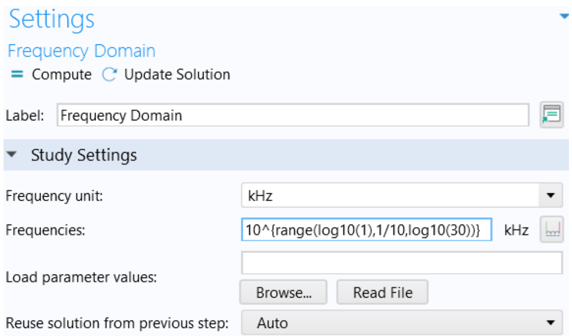

Figure A.2: Settings of the bulk deformation study

For this study, the effective displacement is calculated in COMSOL using the area integration option in the results section. In a similar way, the centre of mass motion of the test mass is determined by volumetric integration of the entire geometry.

This is done by first creating two Global Variables. One for the radius `r_b =sqrt(x^2+y^2)` and the other for the main beam intensity `I_b =exp(2*r_b^2/w0^2)`.

Under Derived Values, the effective deformation is directly calculated as:

```
int_surf(w *I_b) / int_surf(I_b)
```

and the centre of mass motion as:

```
int_vol(w ) / int_vol(1)
```

where `int_vol` and `int_surf` are volumetric and surface integration variables defined in the Definitions section.

B Photon calibrator

In this appendix, additional calculations, figures and concepts related to the photon calibration system are provided. All code files used for and referenced in this thesis can be found at `https://github.com/larskooy01/master-thesis`.

B.1 Force imbalance due to mirror curvature

The test mass has a radius of curvature of approximately 14.5 m. The force direction of a reflected beam is normal to the surface, which means that the two offset Pcal beams each have a slightly different direction, as visualised in the figure below.

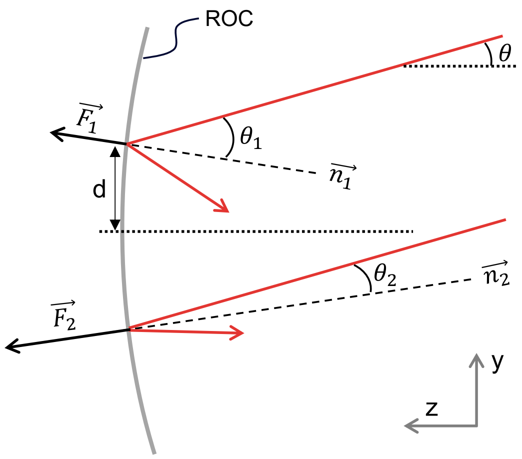

Figure B.1: Visualisation of the two force directions (not to scale)

The error on the total induced force F is given by:

$$e_F = \frac{F_{flat}}{F_{concave}} - 1 = \frac{2\cos\theta}{\cos\theta_1\,\cos\theta_{n_1} + \cos\theta_2\,\cos\theta_{n_2}} - 1 \tag{B.1}$$

Although the forces F_1 and F_2 individually have an error of ~0.3 %, the total force has a negligible error of less than 0.001% for a distance of $d = 45$ mm. This relative error could induce a rotational error, since the force imbalance creates a moment M_x about the x-axis.

$$M_x = d\,F_1 - d\,F_2 \tag{B.2}$$

with the offset of the centre of force Δy given by

$$\Delta y = d\,\frac{\cos(\theta + d/R) - \cos(\theta - d/R)}{\cos(\theta + d/R) + \cos(\theta - d/R)} = -d\,\frac{\sin(d/R)\sin\theta}{\cos(d/R)\cos\theta} \approx -\frac{d^2}{R}\tan\theta = 0.12 \quad \text{mm} \tag{B.3}$$

From the rotational error plot in Figure 3.7, it is found that this is expected to result in an calibration error of less than 0.01 %.

B.2 Elliptical spot size effect

For a Pcal beam at an angle of incidence θ, the incident spot forms an ellipse rather than a circle. Assuming the Pcal beam path is in the yz-plane, the major axis of the elliptical spot has waist radius of $w_2 = w_1/\cos\theta$. It is investigated whether this has an effect on the effective displacement.

Using the COMSOL models presented in Appendix section A.2 and A.3, the results are compared considering various angles of incidence. Where, in the case of the two beam model, the load is modelled as:

$$\texttt{-p* (exp(-2*(x}^2\texttt{/w1}^2\texttt{+(y-d)}^2\texttt{/w2}^2\texttt{)) + exp(-2*(x}^2\texttt{/w1}^2\texttt{+(y+d)}^2\texttt{/w2}^2\texttt{)))}$$

Two variations of the elliptical spot are considered; when the two beam are vertically offset (major y-axis) and horizontally offset (major x-axis). An angle of incidence of 35° is used, and the resulting effective displacement is presented in the table below

Table B.1: Effective displacement for different spot shapes

Spot shape	**Effective displacement**
Circular	$2.204 \cdot 10^{-10}$
Elliptical y	$2.205 \cdot 10^{-10}$
Elliptical x	$2.204 \cdot 10^{-10}$

The difference in effective displacement is negligible, and the resulting variation of the error is even smaller. The greatest difference is observed when the spots are near the edge of the main beam spot (r=2.2 mm). However, at this offset, the split beams have an absolute effective displacement that is too large.

B.3 Power modulation - experimental setup

In the figure below, the experimental setup is shown that was used for the experiment on the SLD power modulation.

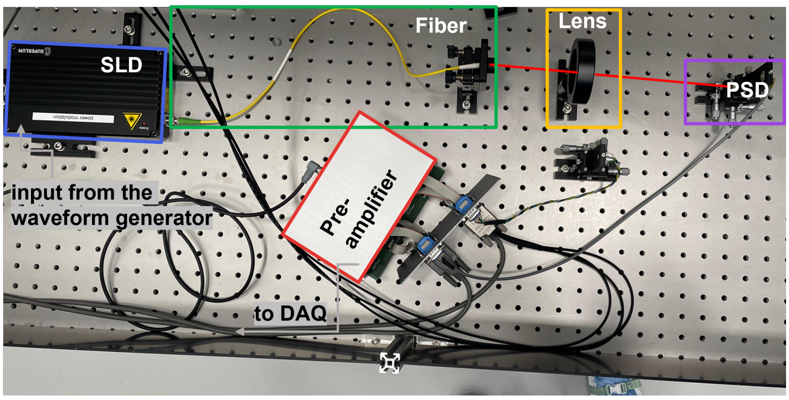

Figure B.2: Experimental setup for modulation of the SLD output power

B.4 Viewport AR coating

AR coating reflectance of the currently installed optical lever viewports at ETpathfinder, as provided by the supplier.

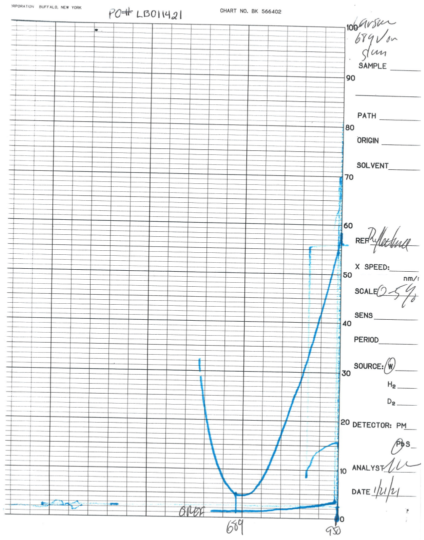

Figure B.3: Reflectance of the viewport AR coating; every 10 ticks is 0.5 %

B.5 Other beam path concepts

Below are some of the other Pcal beam path concepts that have been considered.

Without additional optics

These simplest and most effective concepts were considered first. However, these were quickly found to be infeasible or to lead to large calibration errors.

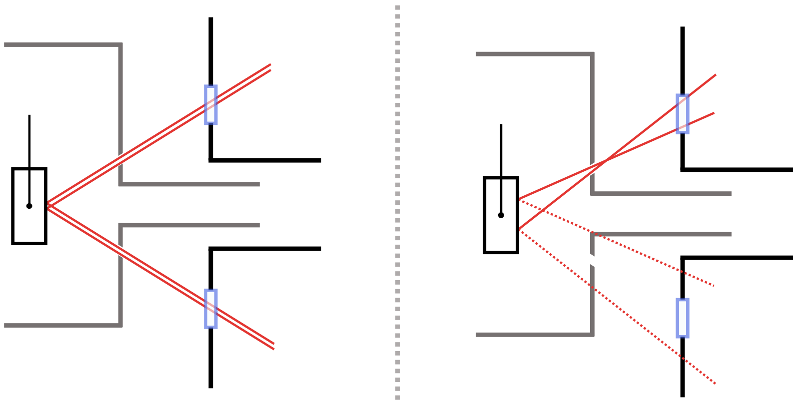

Figure B.4: Beam paths through the shield apertures

Folded beam path using retroreflectors

This concept utilises two co-located transmitters and receivers. Here, the two beam angles and positions are independently controlled. This concept was not chosen because it would require altering and shifting the shield apertures, which would make the optical lever system unfunctional.

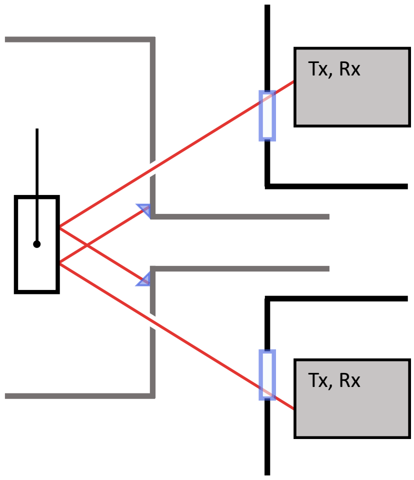

Figure B.5: two retroreflectors to fold the beam path onto itself

C Cryogenic windows

In this appendix, additional results, figures, and calculations related to the cryogenic window model are presented. The complete model, along with other code, can be found at: `https://github.com/larskooy01/master-thesis`.

C.1 Validation of the radiation model

In the tables below, results are shown comparing the radiative heat loads through the apertures for the analytical and numerical models at an aperture size of 20 mm.

Table C.1: Analytical and numerical values of the view factors (D = 20 mm)

View factor	Matlab value [-]	COMSOL value [-]
F_{aa}	1	0.99
F_{ab}	0.037	0.037
F_{ac}	0.0094	0.0093
F_{ad}	0.0049	0.0048

Table C.2: Analytical and numerical values of ambient and mutual radiation (D = 20 mm)

	Ambient radiation [mW]		Mutual radiation [mW]	
Radiation	**Matlab value**	**COMSOL value**	**Matlab value**	**COMSOL value**
Q_a	144	138	0	0
Q_b	5.4	4.9	66	50
Q_c	1.4	1.3	2.5	2.4
Q_d	0.71	0.66	0.56	0.51

C.2 Transmission range for various materials

The figure below from [32], and the information on the webpage [33] are used to select several suitable window materials

Table C.3: Transmissivity of borosilicate for different radiation temperatures

T_r	300 K	250 K	80 K	40 K
τ	0.0007	0.0001	0	0

Table C.4: Transmissivity of c-quartz at different window and radiation temperatures

$T_{window} \backslash T_{rad}$	300 K	250 K	80 K	40 K
250 K	0.02	0.03	0.32	0.61
80 K	0.04	0.06	0.44	0.71
40 K	0.05	0.07	0.48	0.74
15 K	0.06	0.09	0.54	0.77

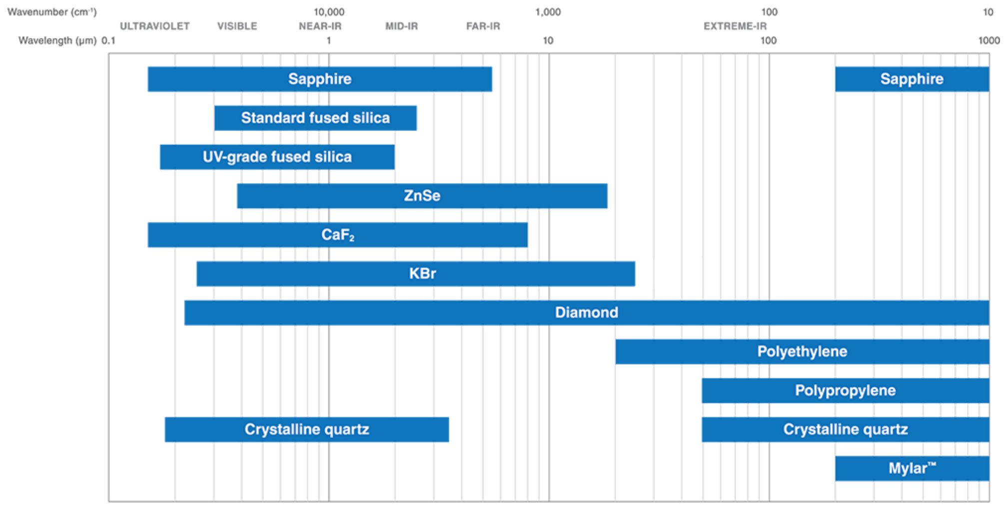

Figure C.1: Wavelength transmission ranges for standard window materials [32]

Figure C.2 shows the spectra of the transmitted radiation when fused silica is used as the window material. When a window is placed in the outer 250 K shield irradiated by the environment at 300 K, it is called a *hot window*. On the other hand, when a window is placed within one of the inner shields, it is called a *cold window* and will be irradiated by different wavelengths of radiation.

Transmitted spectra – fused silica

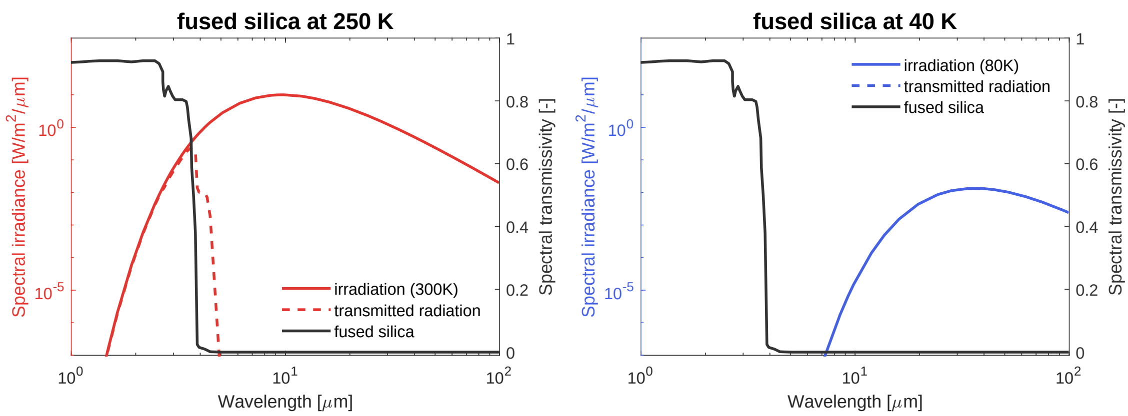

Figure C.2: Spectra of the incident and transmitted radiation when fused silica is used as a hot window (left) and cold window (right)

Transmitted spectra – sapphire

Figure C.3 shows the transmitted spectra for sapphire as a hot and cold window.The peak of the spectral irradiance at 300 K is absorbed mostly by the hot window. At lower temperatures, however, the peak shifts to longer wavelengths, while the transmission band shifts to lower wavelengths. This results in higher transmissivity when used as a cold window

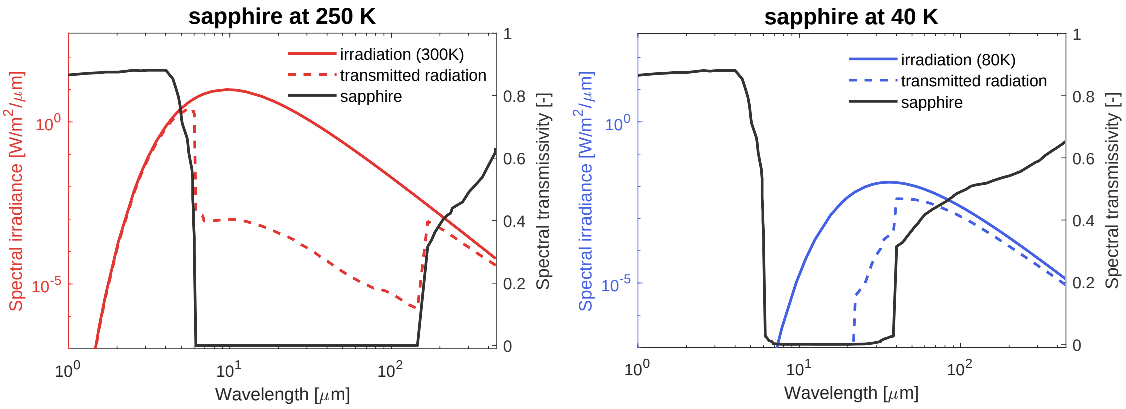

Figure C.3: Spectra of the incident and transmitted radiation when sapphire is used as a hot window (left) and cold window (right)

C.3 Validation of the conduction model

The analytic solution for the radial temperature $T(r)$ is validated against a simulated COMSOL model. In this model, a window within the 80 K shield is considered, subject to a uniform heat load of $Q = 2.16$ W, corresponding to the radiation incident on the window. The thermal conductivities k in W/m K at 80 K for N-BK7 (0.1), fused silica (0.4), and sapphire (700) are used as varying inputs. The thermal contact conductance $h_c = \infty$, such that the edge is cooled to the shield temperature.

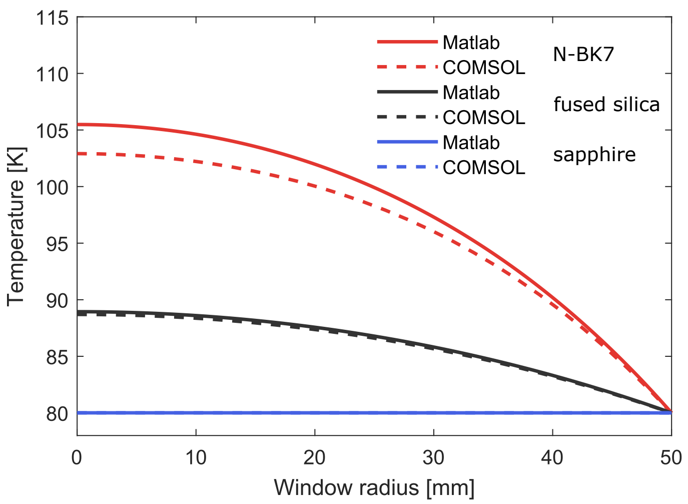

Figure C.4: Radial temperature profile of the analytic and numerical solution

In Figure C.4, the analytic solutions with a relatively small temperature gradient ($\Delta T < 10$ K) accurately follow the numerical solution. At large temperature gradients ($\Delta T > 20$ K), the analytic solution overestimates the window temperature. This is due to the linearised h_{rad} term, since the true radiative cooling scales with T^4. At higher temperatures, this leads to a smaller increase in temperature than in the linearised radiation case.

C.4 Derivation of the radial window temperature

Thin circular window of radius R, thickness t, and thermal conductivity k. The window is subject to a uniform total heat load Q in W, and it loses heat by radiation on both faces to the environment at temperature T_0. The circular edge at $r = R$ is cooled conductively through a thermal contact conductance h_c to the same environment at T_0.

Assumptions

- axisymmetry,
- steady state,
- uniform temperature through the thickness (thin window approximation).

Heat equation

$$q = \frac{Q}{\pi R^2} \quad [\mathrm{W/m^2}]. \tag{C.1}$$

$$q_v = \frac{q}{t} = \frac{Q}{\pi R^2 t} \quad [\mathrm{W/m^3}]. \tag{C.2}$$

Radiation on both faces by a linearised radiation coefficient h_{rad} in W/(m^{2}K), so that

$$q_{\mathrm{rad}} = h_{\mathrm{rad}}\big(T - T_0\big). \tag{C.3}$$

$$h_{\mathrm{rad}} \approx 2 \cdot 4\varepsilon\sigma T_0^3 = 8\varepsilon\sigma T_0^3, \tag{C.4}$$

The axisymmetric steady-state conduction equation with radiative cooling:

$$\frac{1}{r}\frac{d}{dr}\left(rk\frac{dT}{dr}\right) - \frac{h_{\mathrm{rad}}}{t}\big(T - T_0\big) + q_v = 0. \tag{C.5}$$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) - \lambda^2\big(T - T_0\big) = -\frac{Q}{\pi R^2 kt}, \tag{C.6}$$

where we define

$$\lambda^2 = \frac{h_{\mathrm{rad}}}{kt}. \tag{C.7}$$

Homogeneous and particular solutions

$$\Theta(r) = T(r) - T_0. \tag{C.8}$$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\Theta}{dr}\right) - \lambda^2\Theta = -\frac{Q}{\pi R^2 kt}. \tag{C.9}$$

$$\frac{d^2\Theta}{dr^2} + \frac{1}{r}\frac{d\Theta}{dr} - \lambda^2\Theta = -\frac{Q}{\pi R^2 kt}. \tag{C.10}$$

The homogeneous equation is

$$\frac{d^2\Theta}{dr^2} + \frac{1}{r}\frac{d\Theta}{dr} - \lambda^2\Theta = 0, \tag{C.11}$$

with a general solution

$$\Theta_h(r) = C_1 I_0(\lambda r) + C_2 K_0(\lambda r), \tag{C.12}$$

where I_0 and K_0 are the modified Bessel functions of the first and second kind of order zero, and C_1, C_2 are constants.

$$C_2 = 0. \tag{C.13}$$

$$\Theta_h(r) = C_1 I_0(\lambda r). \tag{C.14}$$

(C.9) reduces to

$$-\lambda^2\Theta_p = -\frac{Q}{\pi R^2 kt}, \tag{C.15}$$

$$\Theta_p = \frac{Q}{\pi R^2 kt\lambda^2}. \tag{C.16}$$

$$\lambda^2 = \frac{h_{\text{rad}}}{kt} \quad \Rightarrow \quad kt\lambda^2 = h_{\text{rad}}, \tag{C.17}$$

$$\Theta_p = \frac{Q}{\pi R^2 h_{\text{rad}}}. \tag{C.18}$$

$$T_p = T_0 + \frac{Q}{\pi R^2 h_{\text{rad}}}. \tag{C.19}$$

The general solution for $T(r)$

$$T(r) = T_0 + C_1 I_0(\lambda r) + \frac{Q}{\pi R^2 h_{\text{rad}}}. \tag{C.20}$$

Boundary conditions

BC 1: Symmetry at the centre At $r = 0$, axisymmetry implies

$$\left.\frac{dT}{dr}\right|_{r=0} = 0. \tag{C.21}$$

This is automatically satisfied by $I_0(\lambda r)$, since I_0 is even and $I_1(0) = 0$.

BC 2: Edge conduction cooling At the edge $r = R$, the window is cooled conductively through a contact conductance h_c to the environment at T_0. This gives a Robin boundary condition:

$$-k \left.\frac{dT}{dr}\right|_{r=R} = h_c\big(T(R) - T_0\big). \tag{C.22}$$

From the general solution (C.20)

$$T(R) - T_0 = C_1 I_0(\lambda R) + \frac{Q}{\pi R^2 h_{\text{rad}}}. \tag{C.23}$$

$$\frac{dT}{dr} = C_1 \lambda I_1(\lambda r), \tag{C.24}$$

$$\left.\frac{dT}{dr}\right|_{r=R} = C_1 \lambda I_1(\lambda R). \tag{C.25}$$

$$-k\lambda C_1 I_1(\lambda R) = h_c \left(C_1 I_0(\lambda R) + \frac{Q}{\pi R^2 h_{\text{rad}}} \right). \tag{C.26}$$

Solving for C_1

$$-k\lambda C_1 I_1(\lambda R) - h_c C_1 I_0(\lambda R) = h_c \frac{Q}{\pi R^2 h_{\text{rad}}}, \tag{C.27}$$

$$C_1 \left[-k\lambda I_1(\lambda R) - h_c I_0(\lambda R)\right] = h_c \frac{Q}{\pi R^2 h_{\text{rad}}}, \tag{C.28}$$

$$C_1 = -\frac{h_c}{k\lambda I_1(\lambda R) + h_c I_0(\lambda R)} \frac{Q}{\pi R^2 h_{\text{rad}}}. \tag{C.29}$$

Final expression

Inserting C_1 (C.29) into (C.20), we get

$$T(r) = T_0 + \frac{Q}{\pi R^2 h_{\text{rad}}} \left[1 - \frac{h_c I_0(\lambda r)}{k\lambda I_1(\lambda R) + h_c I_0(\lambda R)} \right], \tag{C.30}$$

C.5 Window configurations

In this section, other feasible window configurations are presented.

250 K – 80 K – 15 K

This configuration of three fused silica windows, as shown in the figure below, is also expected to meet the desired heat load.

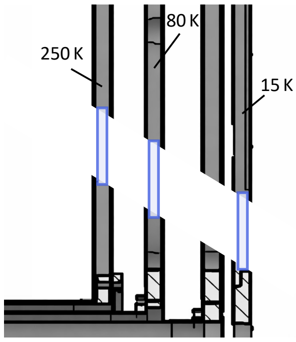

Figure C.5: Heat load Q_d as a function of the contact conductance for various thicknesses of an 80 K outer window and a 40 K inner window

Compared to the 80 K – 40 K – 15 K configuration, this configuration requires a slightly higher minimum thermal contact conductance. It is not selected as the proposed configuration, however, it is still expected to be able to reach the target heat load with a relatively low h_c.

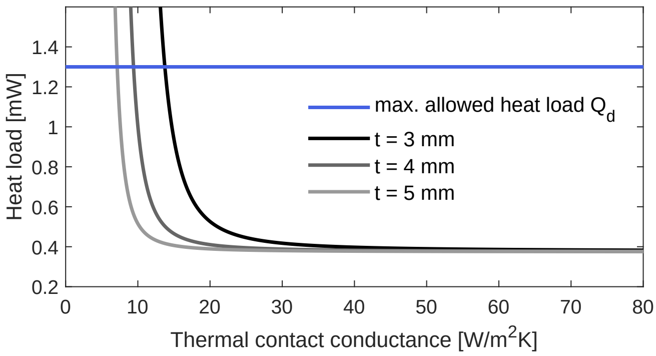

Figure C.6: Heat load Q_d as a function of the contact conductance for various thicknesses of an 80 K outer window and a 40 K inner window

C.6 Window cooling methods

It has been shown that sufficient cooling of the windows is crucial for reducing the heat load to several mW. In this section, a qualitative study of various methods for cooling the windows is presented. Using numerical modelling, the effects of having an increased r non-uniform contact surface are investigated.

Cooling by clamping

The variable c is the radial distance to the added window. This area is clamped at either side of the window by the shield or mount, which is considered to be at the shield temperature of 80 K. The outward radiative heat is the total heat in W, radiated by one face of the window. This is plotted in the figure below.

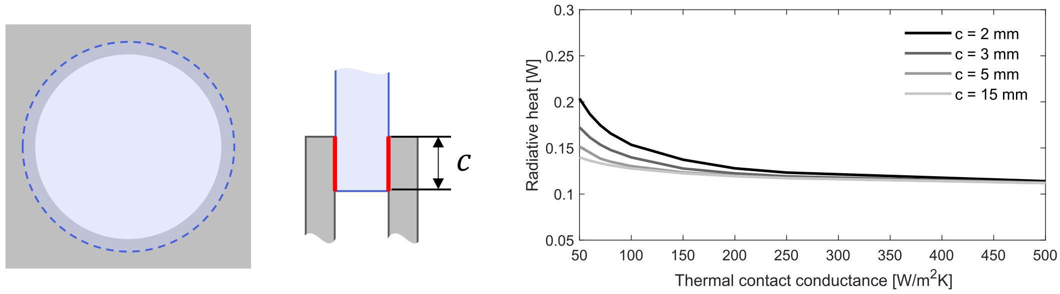

Figure C.7: Outgoing radiative heat as a function of the contact conductance for various amounts of cooling by clamping (window thickness is 2 mm)

Non-uniform cooling

The variable e denotes the fraction of the window circumference that is cooled, assuming three-fold symmetric cooling, as depicted on the left in the figure below. The edge of the window is cooled to the shield temperature of 80 K. The outward radiative heat is the total heat in W, radiated by one face of the window. This is plotted in the figure below.

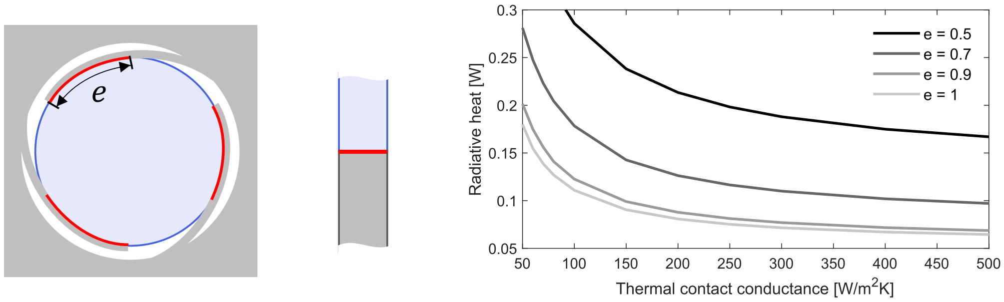

Figure C.8: Outgoing radiative heat as a function of the contact conductance for various amounts of asymmetric cooling at the edge (window thickness is 3 mm)